EDA

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Clean the Data

```r
singapore <- read.csv('BirthsAndFertilityRatesAnnual.csv')
```

```r
clean_1 <- singapore |>
  filter(DataSeries %in% c("Total Fertility Rate (TFR)", "Total Live-Births"))
```

```r
clean_2 <- pivot_longer(clean_1,
                        cols = starts_with("X"),
                        names_to = "Year",
                        values_to = "Value",
                        values_transform = list(Value = as.character))
```

```r
clean_3 <- mutate(clean_2, Year = str_remove(Year, "X"))

clean_4 <- clean_3 |> mutate(Year = as.numeric(Year), Value = as.numeric(Value))

clean_data <- clean_4 |>
  pivot_wider(names_from = DataSeries, values_from = Value)
singapore <- clean_data |>
  as_tsibble(index = Year)
singapore
```

```
## # A tsibble: 66 x 3 [1Y]
##     Year `Total Fertility Rate (TFR)` `Total Live-Births`
##    <dbl>                        <dbl>               <dbl>
##  1  1960                         5.76               61775
##  2  1961                         5.41               59930
##  3  1962                         5.21               58977
##  4  1963                         5.16               59530
##  5  1964                         4.97               58217
##  6  1965                         4.66               55725
##  7  1966                         4.46               54680
##  8  1967                         3.91               50560
##  9  1968                         3.53               47241
## 10  1969                         3.22               44562
## # i 56 more rows
```

The data has been extracted, and the EDA is only on the Total Fertility Rate (TFR) and Total Live-Births (TLB).

```
singapore |> autoplot(`Total Fertility Rate (TFR)`) +
  labs(title = "Total Fertility Rate (1960-2024)",
       caption = "Figure 1. Plot of the Total Fertility Rate in Singapore from 1960 to 2024.",
       y = "Total Fertility Rate",
       x = "Year") +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0)
        )
```

```
## Warning: ‘autoplot.tbl_ts()‘ was deprecated in fabletools 0.6.0.
## i Please use ‘ggtime::autoplot.tbl_ts()‘ instead.
## i Graphics functions have been moved to the {ggtime} package. Please use
##   ‘library(ggtime)‘ instead.
## This warning is displayed once per session.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```

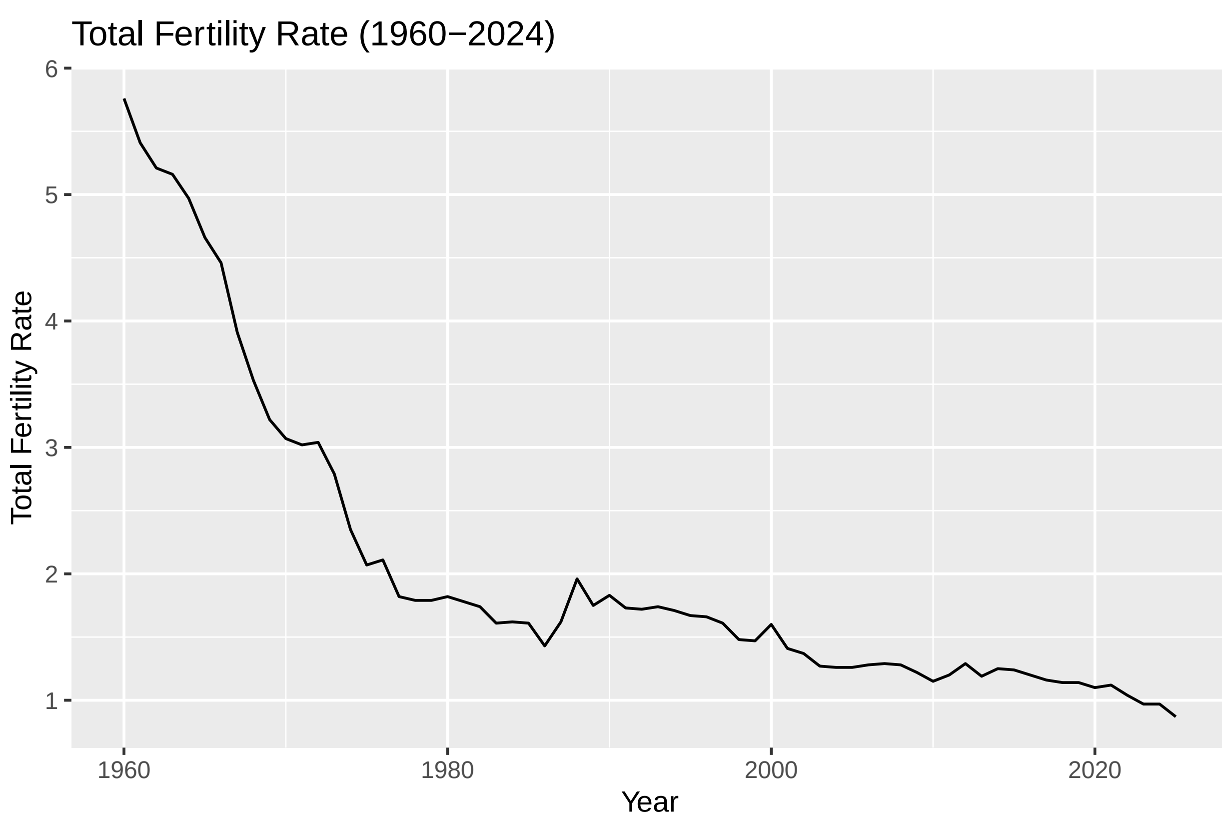


Figure 1. Plot of the Total Fertility Rate in Singapore from 1960 to 2024.

```
singapore |>
  autoplot(`Total Live-Births`) +
  labs(
    title = "Total Live-Births (1960-2024)",
      caption = stringr::str_wrap(
```

```r
        paste0(
          "Figure 2. Plot of the Total Live-Births in Singapore from",
          " 1960 to 2024 shows that it follows a roughly similar trend",
          " as the plot of the Total Fertility Rate."
          ),
        width = 80
        ),
      y = "Total Live-Births",
      x = "Year") +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0)
        )
```

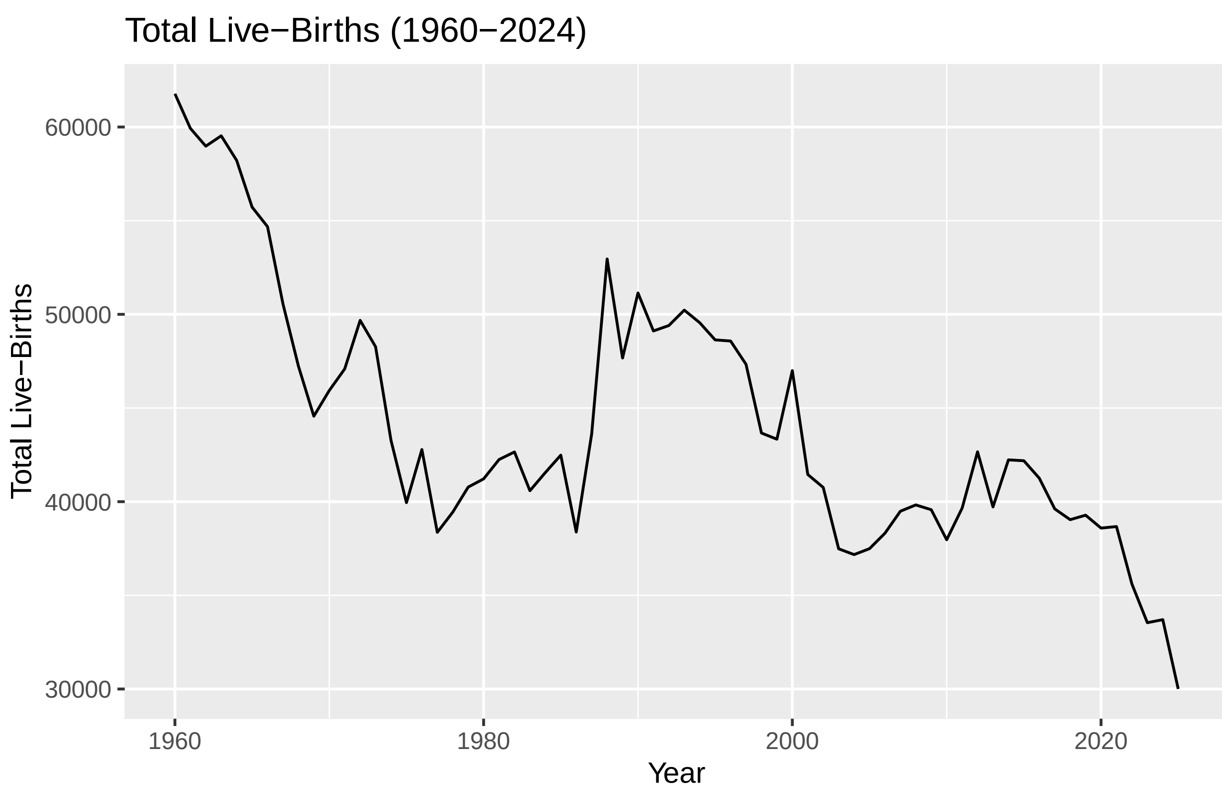


Figure 2. Plot of the Total Live−Births in Singapore from 1960 to 2024 shows that it follows a roughly similar trend as the plot of the Total Fertility Rate.

While figure 1 and figure 2 look different we can see that they show spikes and drops at similar points. There is a spike around 1988 in the plot of Total Fertility Rate and a large spike around the same year in the plot of the Total Live-Births. This can also be seen with a spike around 1972, as well as drops around 1969, 1987, 2003, and 2024.

```r
singapore <- singapore |>
  mutate(
    log_TFR = log(`Total Fertility Rate (TFR)`),
    log_TLB = log(`Total Live-Births`)
  )
```

```
singapore |> autoplot(log_TFR) +
  labs(title = "Log-Transformed Total Fertility Rate (1960-2024)",
       caption = stringr::str_wrap("Figure 3. Plot of the Total Fertility Rate with a log transformatio
       y = "log(Total Fertility Rate)",
       x = "Year") +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0)
        )
```

```
## Ignoring unknown labels:
## * width : "80"
```

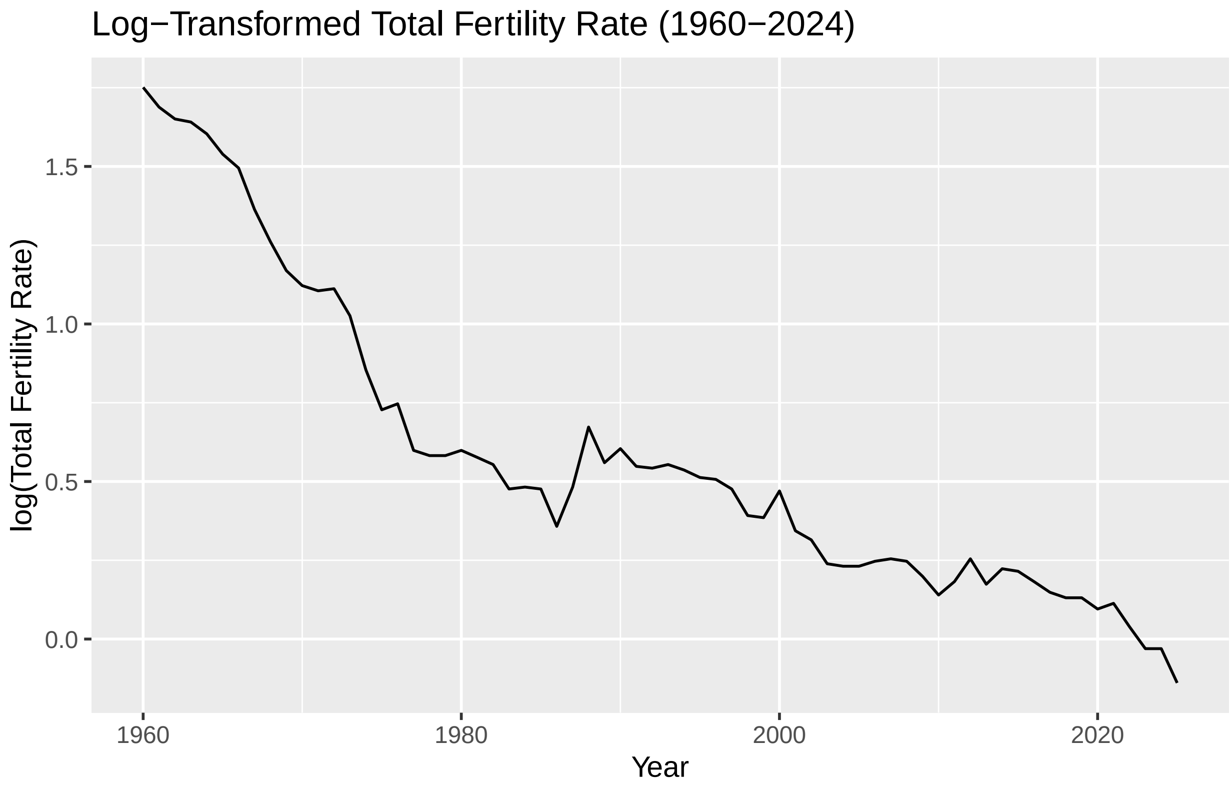


Figure 3. Plot of the Total Fertility Rate with a log transformation in Singapore from 1960 to 2024.

```
singapore |> autoplot(log_TLB) +
  labs(title = "Log-Transformed Total Live-Births (1960-2024)",
       caption = stringr::str_wrap("Figure 4. Plot of the Total Live-Births with a log transformation i
       y = "log(Total Live-Births)",
       x = "Year") +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0)
        )
```

```
## Ignoring unknown labels:
## * width : "80"
```

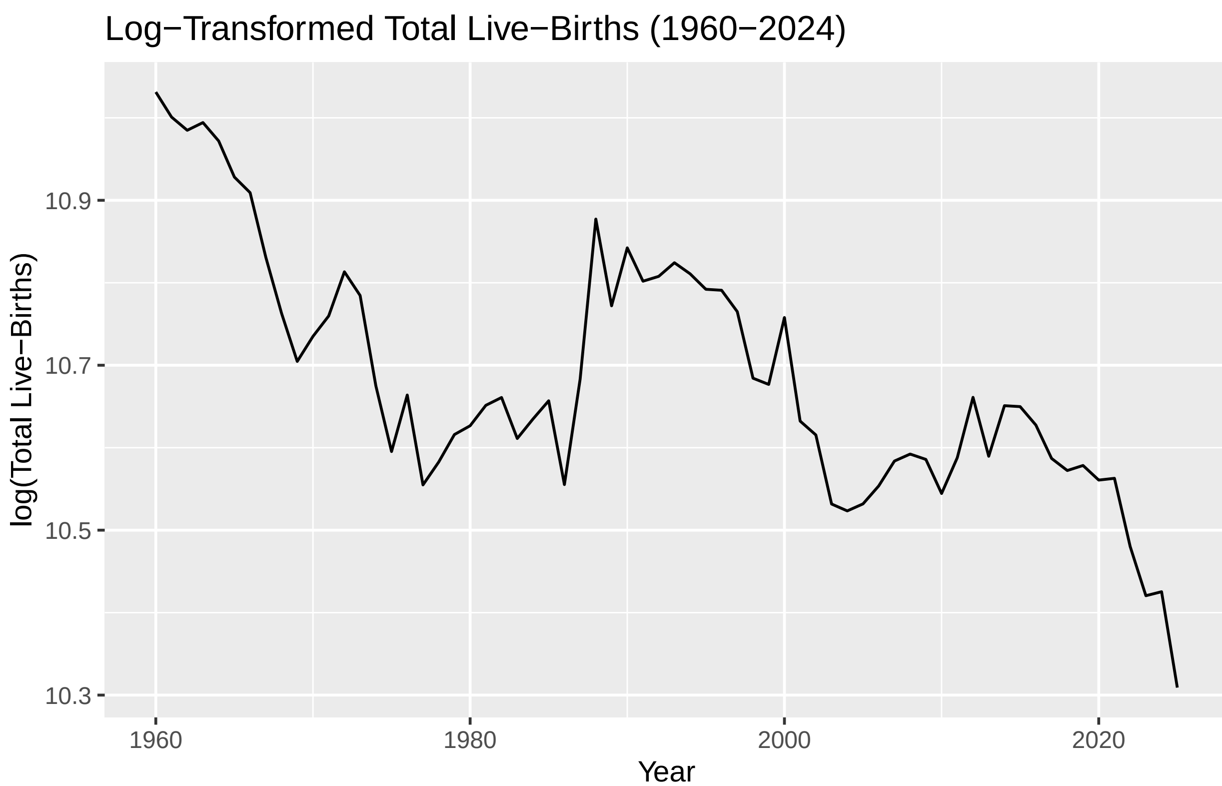


Figure 4. Plot of the Total Live−Births with a log transformation in Singapore from 1960 to 2024.

```
singapore |> features(log_TFR, unitroot_kpss)
```

```
## # A tibble: 1 x 2
##   kpss_stat kpss_pvalue
##       <dbl>       <dbl>
## 1      1.49        0.01
```

Table 1. KPSS test of the log transformed Total Fertility rate in Singapore from 1960-1924. The p-value is 0.01, which is significant at the $\alpha = 0.05$ significance level.

```
singapore |> features(log_TLB, unitroot_kpss)
```

```
## # A tibble: 1 x 2
##   kpss_stat kpss_pvalue
##       <dbl>       <dbl>
## 1     0.996        0.01
```

Table 2. KPSS test of the log transformed Total Live-Birth rate in Singapore from 1960-1924. The p-value is 0.01, which is significant at the $\alpha = 0.05$ significance level.

Figures 3 and 4, along with the p-values from tables 1 and 2 both being significant at the 0.05 significance level, show that a log transformation seems to help a bit but they both still need to be stabilized more. To do this, they will be differenced.

```r
singapore |> autoplot(difference(log_TFR, 1)) +
  labs(title = "First-Order difference of Log TFR (1960-2024)",
       caption = stringr::str_wrap("Figure 5. Plot of the first-order difference of the log transformed
       y = "Change in log(TFR)",
       x = "Year") +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0)
        )
```

```
## Ignoring unknown labels:
## * width : "80"
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (‘geom_line()‘).
```

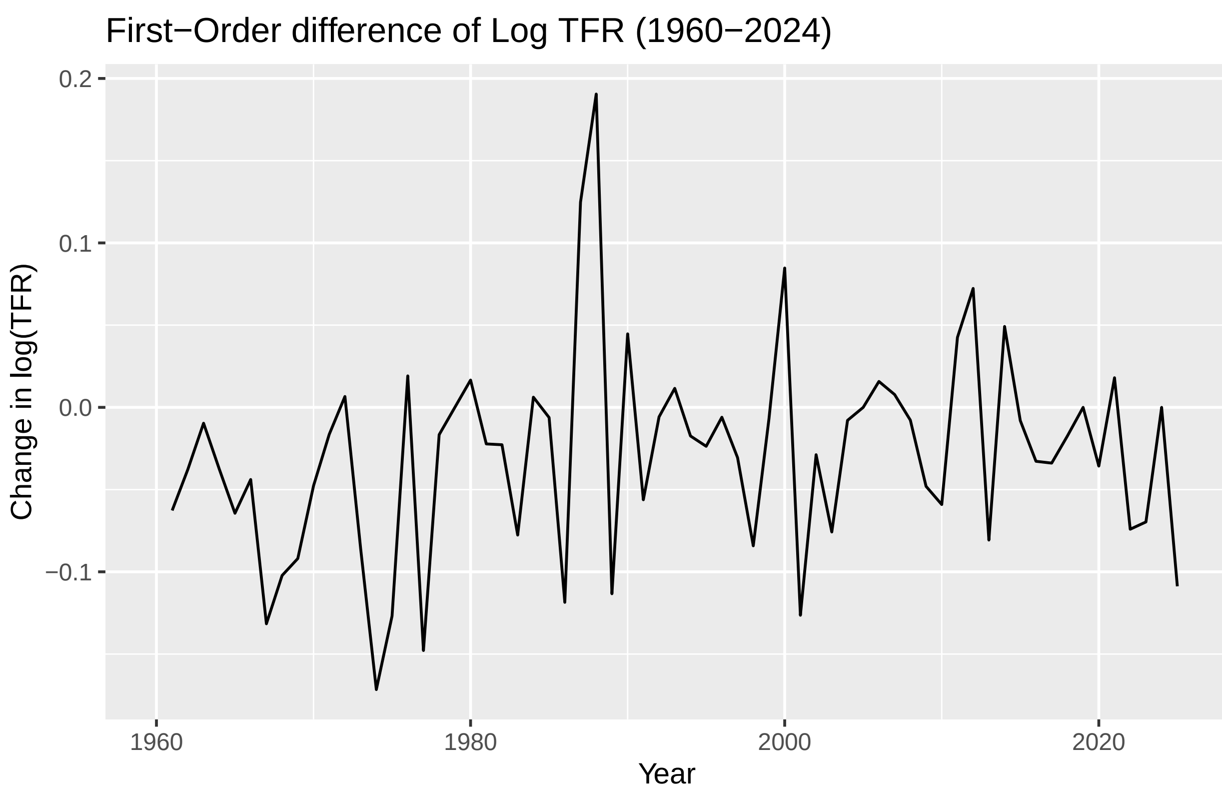


Figure 5. Plot of the first−order difference of the log transformed Total Fertility Rate in Singapore from 1960 to 2024.

```r
singapore |> autoplot(difference(log_TLB, 1)) +
  labs(title = "First-Order difference of Log TLB (1960-2024)",
       caption = stringr::str_wrap("Figure 6. Plot of the first-order difference of the log transformed
       width = 80,
       y = "Change of log(TLB)",
       x = "Year") +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0)
        )
```

```
## Ignoring unknown labels:
## * width : "80"
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_line()`).
```

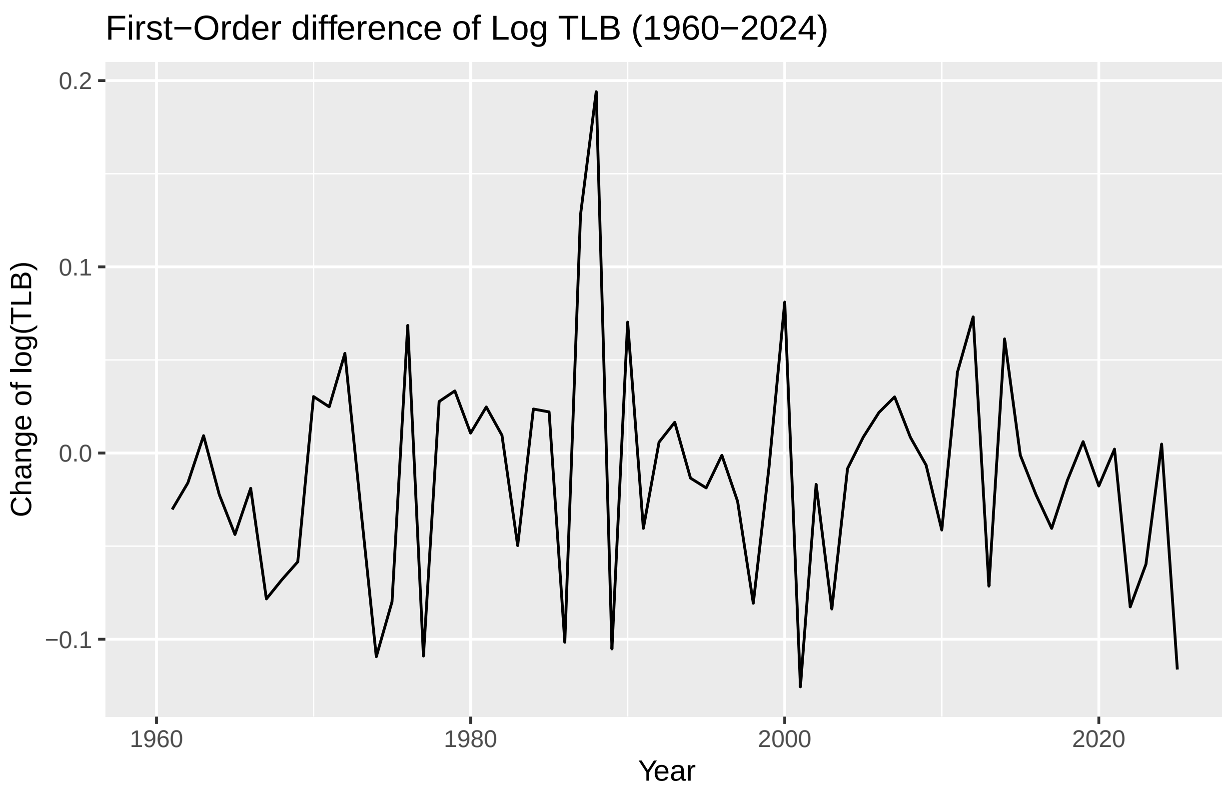


Figure 6. Plot of the first−order difference of the log transformed Total Live−Births in Singapore from 1960 to 2024.

Plots 5 and 6 no longer show a downward trend with a first-order difference of the log transform of each variable.

```r
singapore |> features(difference(log_TFR, 1), unitroot_kpss)
```

```
## # A tibble: 1 x 2
##   kpss_stat kpss_pvalue
##       <dbl>       <dbl>
## 1     0.370      0.0902
```

Table 3. KPSS test of the first-order differenced log transformed Total Fertility rate in Singapore from 1960-1924. The p-value is 0.09, which is not significant at the $\alpha = 0.05$ significance level.

```r
singapore |> features(difference(log_TLB, 1), unitroot_kpss)
```

```
## # A tibble: 1 x 2
##   kpss_stat kpss_pvalue
##       <dbl>       <dbl>
## 1     0.114         0.1
```

Table 4. KPSS test of the first-order differenced log transformed Total Live-Birth rate in Singapore from 1960-1924. The p-value is 0.1, which is not significant at the $\alpha = 0.05$ significance level.

Tables 3 and 4 confirm what figures 5 and 6 showed, when both Total Live-Births and Total Fertility rate are log transformed and then differenced, they are stationary. In a KPSS test, if the p-value is less than 0.05, the null-hypothesis that the series is stationary, is rejected. However, if the p-value is greater than 0.05, the null-hypothesis is accepted and the series is found to be stationary.

```
singapore |> ACF(difference(log_TFR, 1)) |>
  autoplot() +
  labs(title = "ACF of First-Order Differenced Log TFR (1960-2024)",
       caption = stringr::str_wrap("Figure 7. Autocorrelation Function of the stationary Total Fertilit
       width = 80)
       ) +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0)
)
```

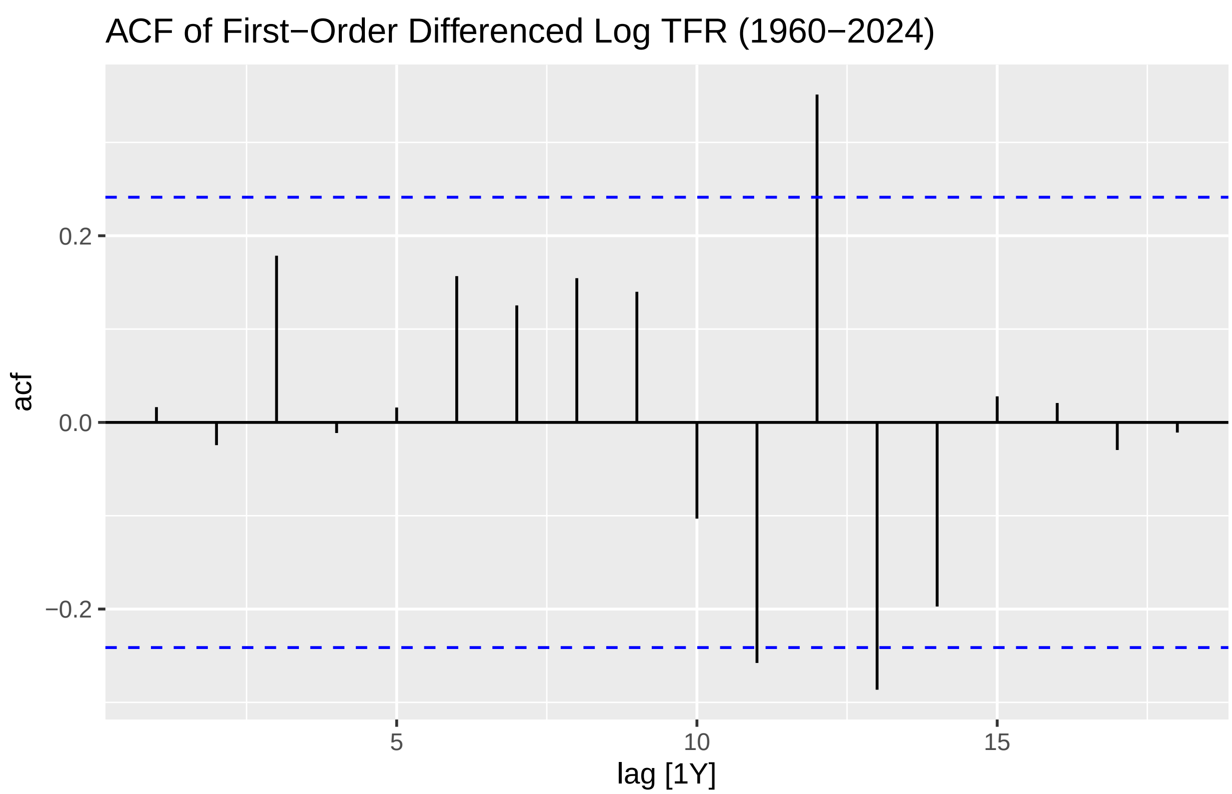


Figure 7. Autocorrelation Function of the stationary Total Fertility Rate data in Singapore from 1960−2024.

```
singapore |> ACF(difference(log_TLB, 1)) |>
  autoplot() +
  labs(title = "ACF of First-Order Differenced Log TLB (1960-2024)",
       caption = stringr::str_wrap("Figure 8. Autocorrelation Function of the stationary Total Live-Bir
  ) +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0)
  )
```

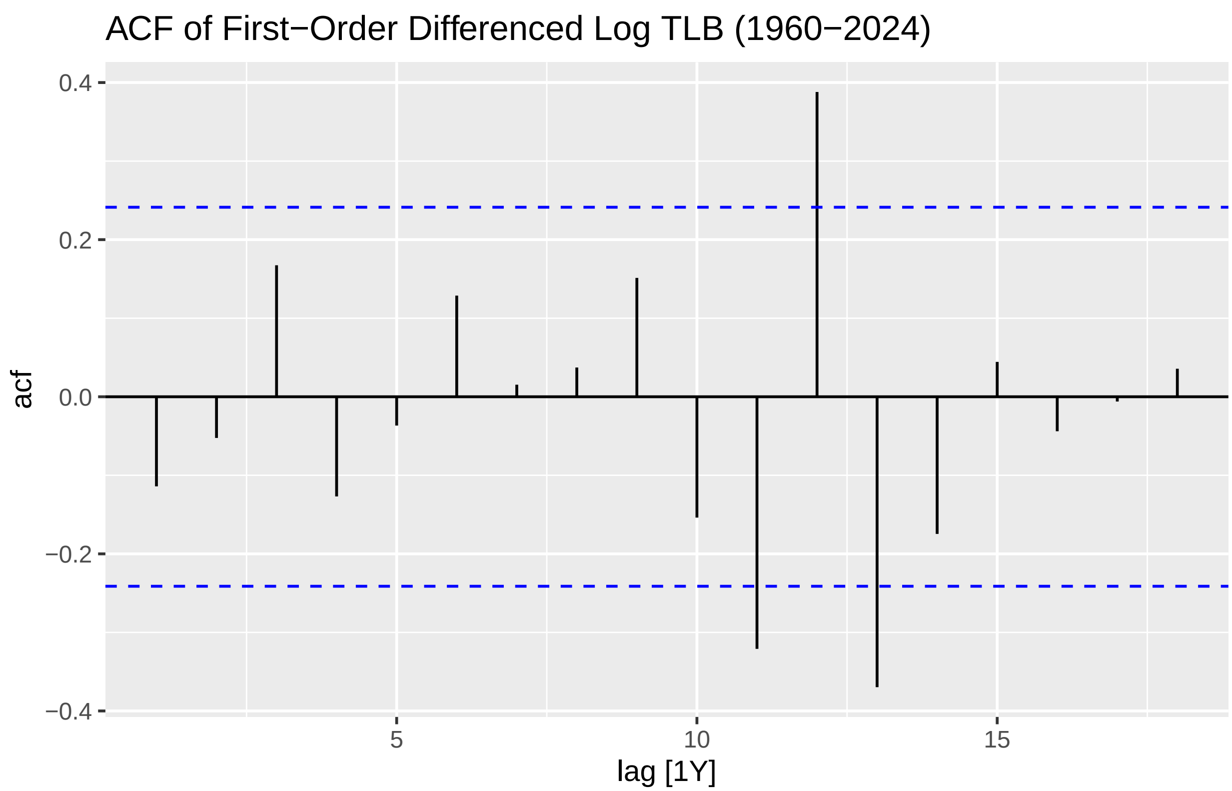


Figure 8. Autocorrelation Function of the stationary Total Live−Births data in Singapore from 1960−2024.

```r
singapore |> PACF(difference(log_TFR, 1)) |>
  autoplot() +
  labs(title = "PACF of First-Order Differenced Log TFR (1960-2024)",
       caption = stringr::str_wrap("Figure 9. Partial Autocorrelation Function of the stationary Total I
  ) +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0)
)
```

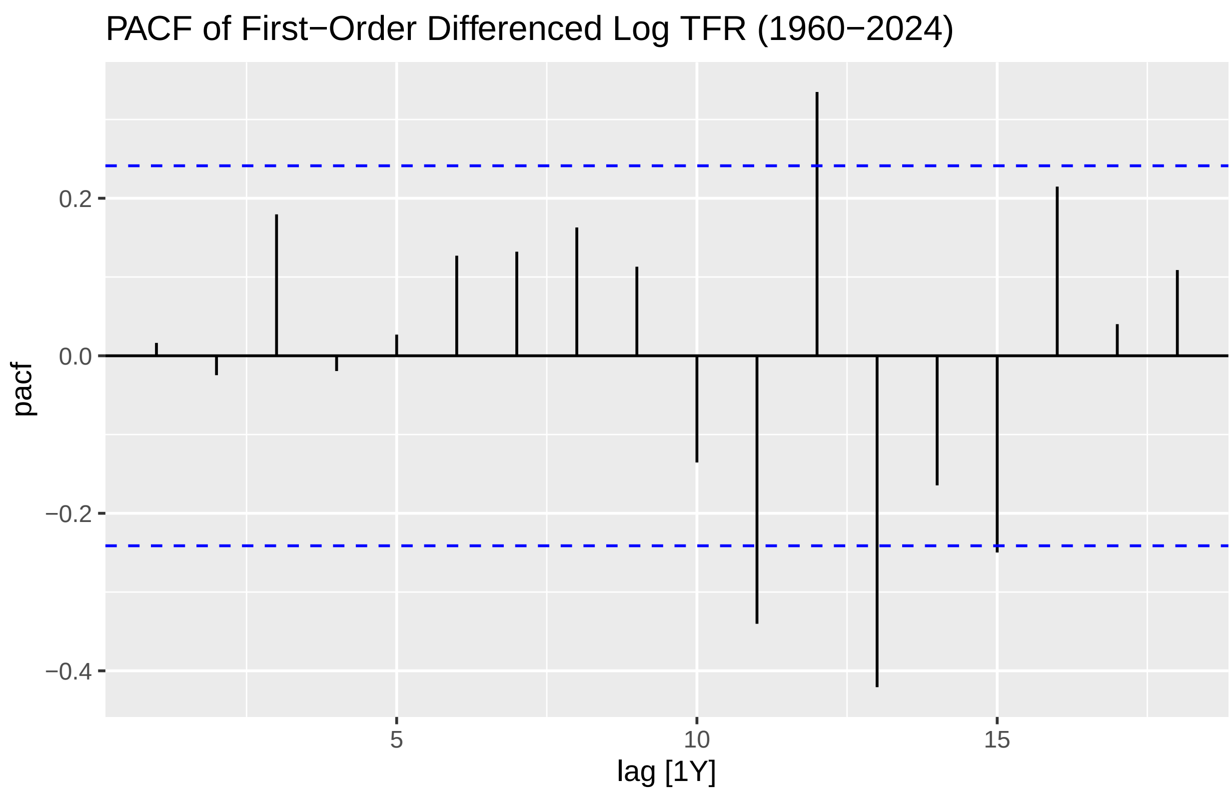


Figure 9. Partial Autocorrelation Function of the stationary Total Fertility
Rate data in Singapore from 1960−2024.

```r
singapore |> PACF(difference(log_TLB, 1)) |>
  autoplot() +
  labs(title = "PACF of First-Order Differenced Log TLB (1960-2024)",
       caption = stringr::str_wrap("Figure 10. Partial Autocorrelation Function of the stationary Total
  ) +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0)
  )
```

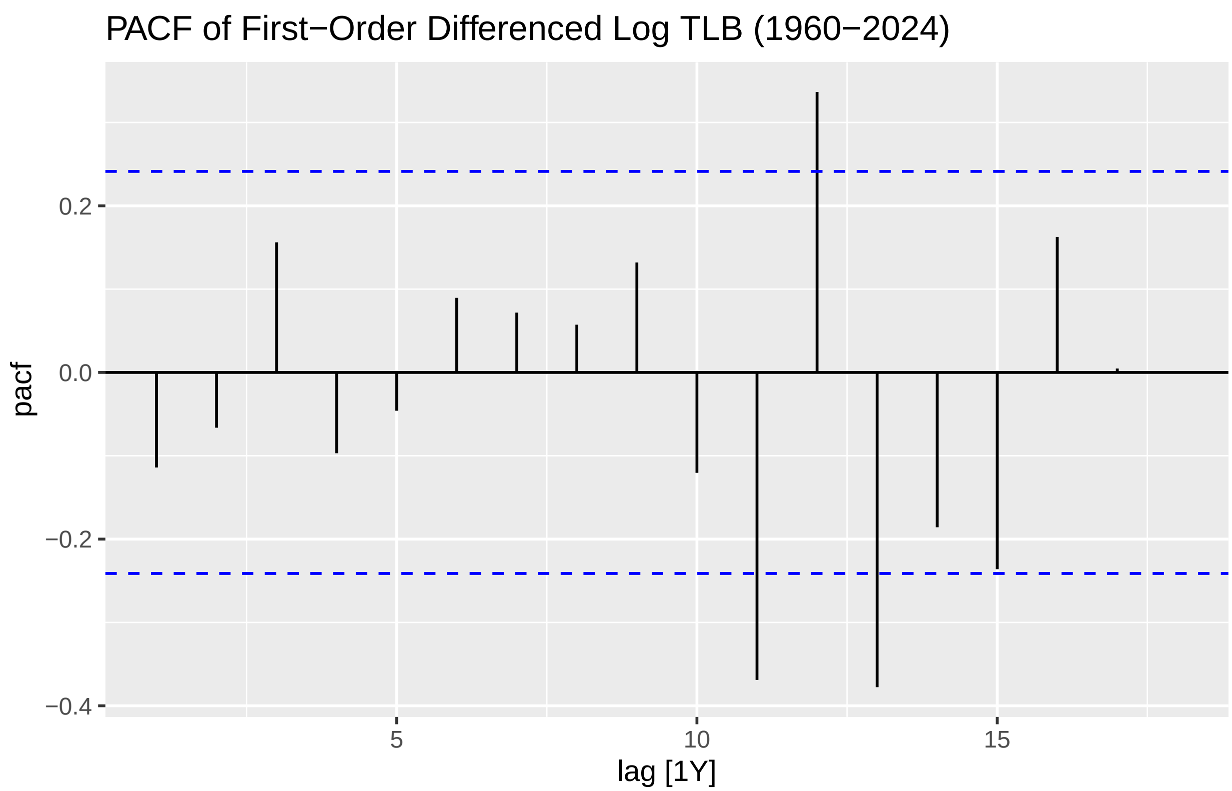


Figure 10. Partial Autocorrelation Function of the stationary Total Live−Births data in Singapore from 1960−2024.

The spikes at lags 11,12,13 in figures 7,8,9, and 10 show that there's a cyclic behaviour. The year of the Dragon historically has had an effect on birth rates in Singapore. Culturally, it is "considered auspicious for births" (The Straits Times). The Year of the Dragon comes every 12 years, which lines up perfectly with the spikes.

```
TFR_Model1 <- singapore |>
  model(
    tfr_baseline = ARIMA(log_TFR ~ pdq(0,1,0)),
    tfr_auto = ARIMA(log_TFR))
TFR_Model1 |> pivot_longer(everything(), names_to = "Model name", values_to = "Orders")
```

```
## # A mable: 2 x 2
## # Key:     Model name [2]
##   `Model name`                     Orders
##   <chr>                           <model>
## 1 tfr_baseline <ARIMA(0,1,0) w/ drift>
## 2 tfr_auto     <ARIMA(0,1,0) w/ drift>
```

Table 5. This comparison table shows that the auto ARIMA model chose for the Total Fertility Rate is the same as the chosen ARIMA model: ARIMA(0,1,0).

The ARIMA model is chosen with this in mind: ARIMA(p,d,q). Where p is the value chosen by looking at immediate lags that spike on the PACF, d is chosen by the differencing, and q is chosen by looking at immediate lags that spike on the ACF.

```r
TFR_Model1 |> report()
```

```
## # A tibble: 2 x 8
##   .model       sigma2 log_lik   AIC  AICc   BIC ar_roots   ma_roots  
##   <chr>         <dbl>   <dbl> <dbl> <dbl> <dbl> <list>     <list>    
## 1 tfr_baseline 0.00392    88.4 -173. -173. -168. <cpl [0]> <cpl [0]>
## 2 tfr_auto     0.00392    88.4 -173. -173. -168. <cpl [0]> <cpl [0]>
```

Table 6. This comparison table of the chosen TFR baseline ARIMA model and the automatically chose TFR ARIMA model confirms that they are identical.

```r
TFR_Model1 |>
  select(tfr_auto) |> gg_tsresiduals() +
  labs(title = "1960-2024 TFR ARIMA (0,1,0) Plots") +
  theme(plot.title = element_text(hjust = 0.5)
       )
```

```
## Warning: `gg_tsresiduals()` was deprecated in feasts 0.4.2.
## i Please use `ggtime::gg_tsresiduals()` instead.
## i Graphics functions have been moved to the {ggtime} package. Please use
##   `library(ggtime)` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

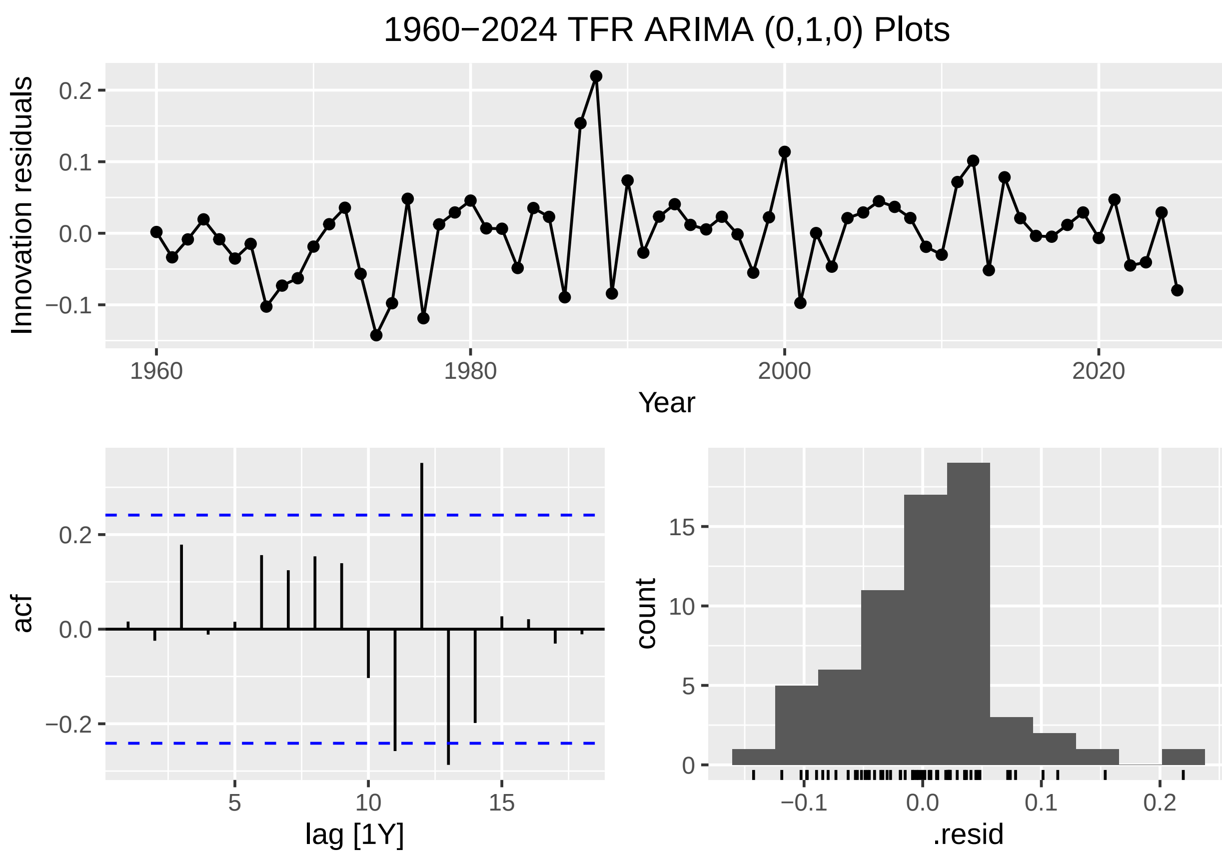


Figure 11. Plots showing the ARIMA(0,1,0) model for Total Fertility Rate in Singapore from 1960-2024. The residuals plot shows the errors bounding roughly around zero. The ACF plot shows spikes at lags 11, 12, and 13. The histogram is unimodal with an abrupt drop off around 0.05 and an outlier around 0.2.

```
TLB_Model <- singapore |>
  model(
    tlb_baseline = ARIMA(log_TLB ~ pdq(0,1,0)),
    tlb_auto = ARIMA(log_TLB))
TLB_Model |> pivot_longer(everything(), names_to = "Model name", values_to = "Orders")
```

```
## # A mable: 2 x 2
## # Key:     Model name [2]
##   `Model name`                  Orders
##   <chr>                        <model>
## 1 tlb_baseline <ARIMA(0,1,0) w/ drift>
## 2 tlb_auto     <ARIMA(0,1,0) w/ drift>
```

Table 7. This comparison table shows that the auto ARIMA model chose for the Total Live-Births is the same as the chosen ARIMA model: ARIMA(0,1,0).

```
TLB_Model |> report()
```

```
## # A tibble: 2 x 8
##   .model        sigma2 log_lik   AIC  AICc   BIC ar_roots   ma_roots
##   <chr>          <dbl>   <dbl> <dbl> <dbl> <dbl> <list>     <list>
## 1 tlb_baseline 0.00348    92.3 -181. -180. -176. <cpl [0]> <cpl [0]>
## 2 tlb_auto     0.00348    92.3 -181. -180. -176. <cpl [0]> <cpl [0]>
```

Table 8. This comparison table of the chosen TLB baseline ARIMA model and the automatically chose TLB ARIMA model confirms that they are identical.

```
TLB_Model |> select(tlb_auto) |> gg_tsresiduals() +
  labs(title = "1960-2024 TLB ARIMA (0,1,0) Plots")
```

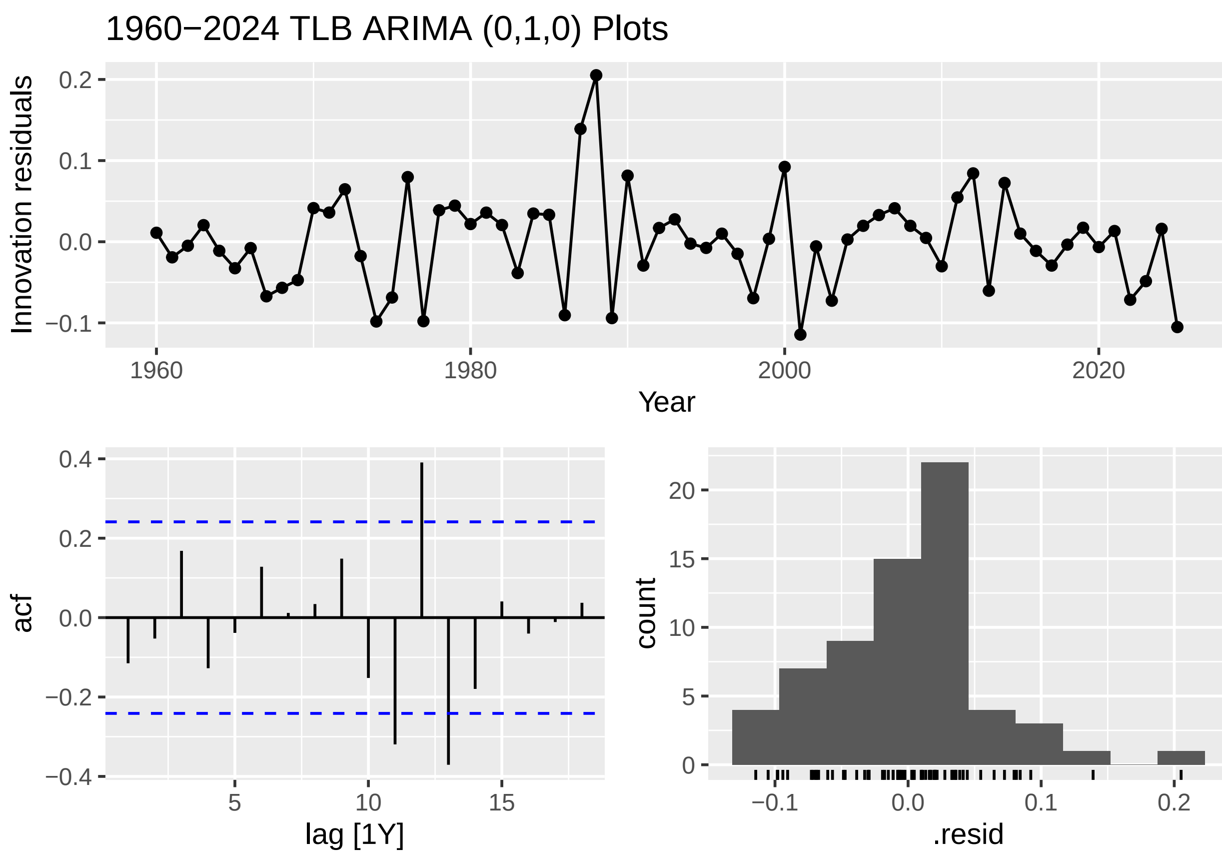


Figure 12. Plots showing the ARIMA(0,1,0) model for Total Live Births in Singapore from 1960-2024. The residuals plot shows the errors bounding roughly around zero. The ACF plot shows spikes at lags 11, 12, and 13. The histogram is unimodal with an outlier around 0.2.

```
singapore_Train <- singapore |>
  filter(Year <= 2012)
```

```
TFR_train_model1 <- singapore_Train |>
  model(tfr_train_baseline = ARIMA(log_TFR ~ pdq(0,1,0)))
TFR_train_forecast <- TFR_train_model1 |>
  forecast(h=12)
TFR_train_forecast |> autoplot() +
  autolayer(singapore, log_TFR) +
  labs(
    title = "ARIMA(0,1,0) Predicted vs Actual TFR",
    caption = stringr::str_wrap(
      "Figure 13. Plot shows that the predictive model aligns very well with the actual data for the 20
    )
  )+
theme(
  plot.title = element_text(hjust = 0.5),
  plot.caption = element_text(hjust = 0)
)
```

```
## Warning: ‘autoplot.fbl_ts()‘ was deprecated in fabletools 0.6.0.
## i Please use ‘ggtime::autoplot.fbl_ts()‘ instead.
```

```
## i Graphics functions have been moved to the {ggtime} package. Please use
##   'library(ggtime)' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

## Warning: 'autolayer.tbl_ts()' was deprecated in fabletools 0.6.0.
## i Please use 'ggtime::autolayer.tbl_ts()' instead.
## i Graphics functions have been moved to the {ggtime} package. Please use
##   'library(ggtime)' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

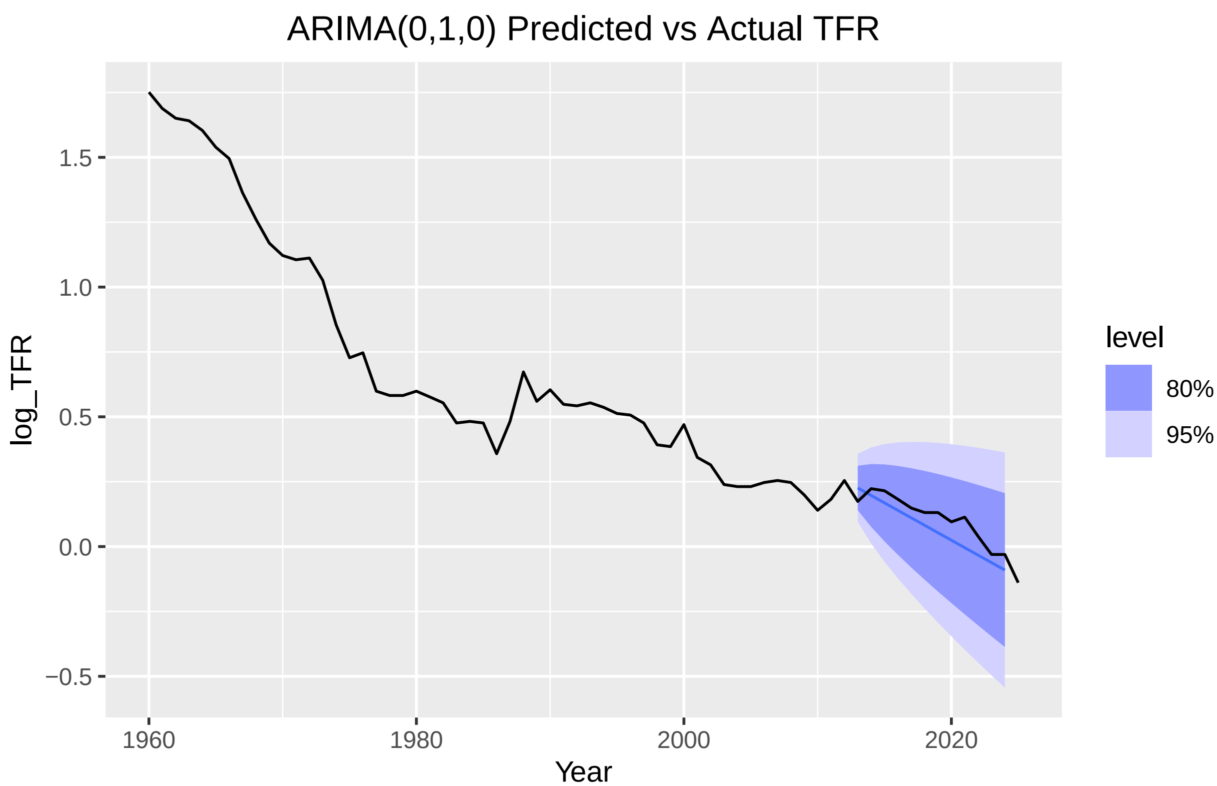


Figure 13. Plot shows that the predictive model aligns very well with the actual data for the 2013−2024 period for the Total Fertility Rate in Singapore.

Figure 13 shows that the predictive Total Fertility Rates in Singapore from 2013-2024 using the ARIMA model line up very well with the actual data. However, the line is linear and has no "wiggle" to it. We can see from the real data how the log of the Total Fertility Rate that while it has an overall downward trend, the data actually wiggles up and down.

```r
TLB_train_model1 <- singapore_Train |>
  model(tlb_train_baseline = ARIMA(log_TLB ~ pdq(0,1,0)))
TLB_train_forecast <- TLB_train_model1 |>
  forecast(h=12)
TLB_train_forecast |> autoplot() +
  autolayer(singapore, log_TLB) +
  labs(
```

```r
    title = "ARIMA(0,1,0) Predicted vs Actual TLB",
    caption = stringr::str_wrap(
      "Figure 14. Plot shows that the ARIMA model predicts a straight line for the log of the TLB in Si
    )
  )+
theme(
  plot.title = element_text(hjust = 0.5),
  plot.caption = element_text(hjust = 0)
)
```

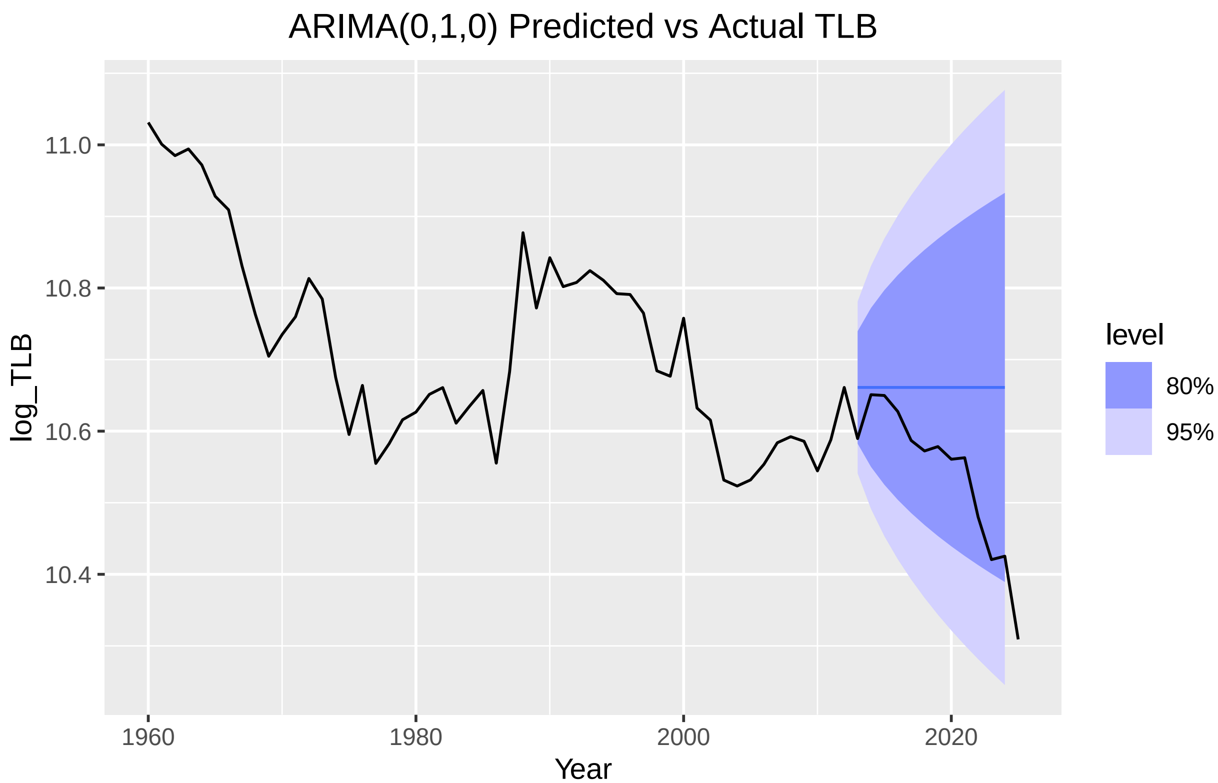


Figure 14. Plot shows that the ARIMA model predicts a straight line for the log of the TLB in Singapore from 2013−2024.

Figures 13 and 14 show that an ARIMA model is not adequate enough at making a predictive modeling the Total Live Births and Total Fertility Rate from 2013-2024 base on the data from 1960-2012.

```r
TFR_Model2 <- singapore |>
  model(
    tfr_baseline = ARIMA(log_TFR ~ pdq(0,1,0)),
    tfr_sarima = ARIMA(log_TFR ~ pdq() + PDQ(period = 12))
  )
TFR_Model2 |> pivot_longer(everything(), names_to = "Model name", values_to = "Orders")
```

```
## # A mable: 2 x 2
## # Key:     Model name [2]
##   `Model name`                                Orders
##   <chr>                                      <model>
## 1 tfr_baseline              <ARIMA(0,1,0) w/ drift>
## 2 tfr_sarima   <ARIMA(0,1,1)(1,0,1)[12] w/ drift>
```

Table 9. Shows the automated search selects a SARIMA(0,1,1)(1,0,1)[12] model with drift for the Total Fertility Rate data.

```r
glance(TFR_Model2)
```

```
## # A tibble: 2 x 8
##   .model       sigma2 log_lik   AIC  AICc   BIC ar_roots    ma_roots
##   <chr>         <dbl>   <dbl> <dbl> <dbl> <dbl> <list>      <list>
## 1 tfr_baseline 0.00392    88.4 -173. -173. -168. <cpl [0]>  <cpl [0]>
## 2 tfr_sarima   0.00292    96.7 -183. -182. -172. <cpl [12]> <cpl [13]>
```

Table 10. Shows the SARIMA(0,1,1)(1,0,1)[12] model for Total Fertility Rate has a lower AIC at around -183, nearly 10 points lower when compared to the ARIMA(0,1,0) model for Total Fertility Rate. The BIC is also lower for the SARIMA model, and the sigma^2 value is also significantly lower for the SARIMA model.

All of the values in table 10 confirm that a seasonal ARIMA model is better at handling the Total Fertility Rate data in Singapore from 1960-2024.

```r
TFR_train_sarima <- singapore_Train |>
  model(TFR_sarima = ARIMA(log_TFR ~ pdq() + PDQ( period = 12)))
TFR_train_sarima
```

```
## # A mable: 1 x 1
##                  TFR_sarima
##                     <model>
## 1 <ARIMA(1,2,1)(1,0,0)[12]>
```

Table 11. Shows that the selected model for the training dataset for Total Fertility Rate in Singapore from 1960-2012 is a SARIMA(1,2,1)(1,0,0)[12] model.

```r
TFR_train_sarima_fc <- TFR_train_sarima |>
  forecast(h = 12)
```

```r
TFR_train_sarima_fc |>
  autoplot() +
  autolayer(singapore, log_TFR) +
  labs(
    title = "SARIMA plot of predicted vs actual TFR",
    caption = stringr::str_wrap(
      "Figure 15. Plot shows that the SARIMA predictive model shows on par fertility rates in singapore
    )
  )+
theme(
  plot.title = element_text(hjust = 0.5),
  plot.caption = element_text(hjust = 0)
)
```

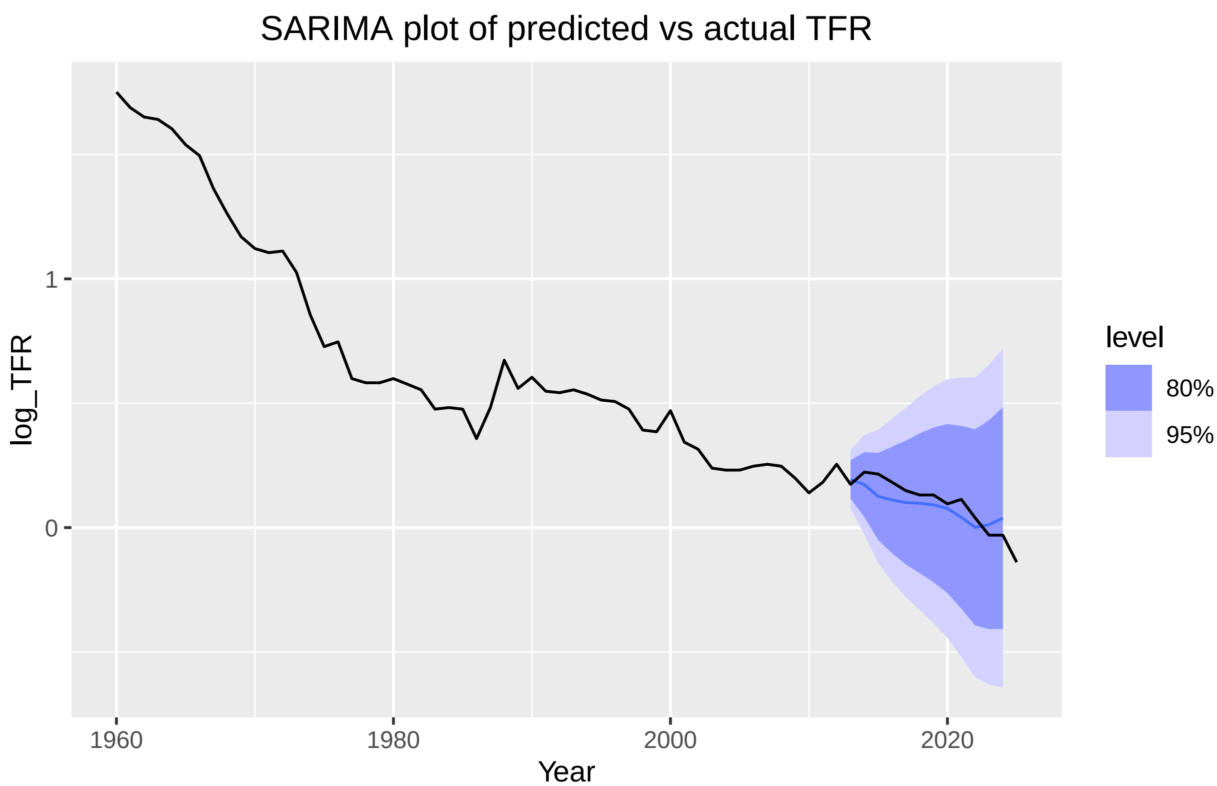


Figure 15. Plot shows that the SARIMA predictive model shows on par fertility rates in singapore from 2013−2024 with the known data.

Figure 15 shows that the SARIMA(0,1,1)(1,0,1)[12] model predicts similar Total Fertility Rates to that of the actual data for Total Fertility Rates in Singapore from 2013-2024. Unlike the ARIMA predictive model seen in figure 13, the SARIMA(0,1,1)(1,0,1)[12] model accounts for seasonality. In the ARIMA model, the predicted values for 2013-2024, was a straight line with a negative slope. The predictive center line for the SARIMA model has "wiggle" to it, accounting for rises and drops in the TFR. This looks much more similar to the actual data. It can be seen that the predictive model curves up at the end, whereas the actual data is down turned.

```
TLB_Model2 <- singapore |>
  model(
    tlb_baseline = ARIMA(log_TLB ~ pdq(0,1,0)),
    tlb_sarima = ARIMA(log_TLB ~ pdq() + PDQ(period = 12))
  )
TLB_Model2 |> pivot_longer(everything(), names_to = "Model name", values_to = "Orders")
```

```
## # A mable: 2 x 2
## # Key:     Model name [2]
##   `Model name`                      Orders
##   <chr>                            <model>
## 1 tlb_baseline   <ARIMA(0,1,0) w/ drift>
## 2 tlb_sarima   <ARIMA(1,1,0)(1,0,0)[12]>
```

Table 12. Shows the automated search selects a SARIMA(1,1,0)(1,0,0)[12] model for the Total Live-Births data.

```r
glance(TLB_Model2)
```

```
## # A tibble: 2 x 8
##   .model        sigma2 log_lik   AIC  AICc   BIC ar_roots   ma_roots
##   <chr>          <dbl>   <dbl> <dbl> <dbl> <dbl> <list>     <list>
## 1 tlb_baseline 0.00348    92.3 -181. -180. -176. <cpl [0]>  <cpl [0]>
## 2 tlb_sarima   0.00274    98.5 -191. -191. -185. <cpl [13]> <cpl [0]>
```

Table 13. Shows the SARIMA(1,1,0)(1,0,0)[12] model for Total Live-Births has a lower AIC at around -191, over 10 points lower when compared to the ARIMA(0,1,0) model for Total Live-Births. The BIC is also lower for the SARIMA model, and the sigma^2 value is also significantly lower for the SARIMA model.

All of the values in table 13 confirm that a seasonal ARIMA model is better at handling the Total Live-Births data in Singapore from 1960-2024.

```r
TLB_train_sarima <- singapore_Train |>
  model(TLB_sarima = ARIMA(log_TLB ~ pdq() + PDQ( period = 12)))
TLB_train_sarima
```

```
## # A mable: 1 x 1
##                  TLB_sarima
##                     <model>
## 1 <ARIMA(0,1,0)(1,0,0)[12]>
```

Table 14. Shows that the selected model for the training dataset for Total Live-Births in Singapore from 1960-2012 is a SARIMA(0,1,0)(1,0,0)[12] model.

```r
TLB_train_sarima_fc <- TLB_train_sarima |>
  forecast(h = 12)
```

```r
TLB_train_sarima_fc |>
  autoplot() +
  autolayer(singapore, log_TLB) +
  labs(
    title = "SARIMA plot of predicted vs actual TLB",
    caption = stringr::str_wrap(
      "Figure 16. Plot shows that the SARIMA predictive model shows similar levels for the total live-b
    )
  )+
theme(
  plot.title = element_text(hjust = 0.5),
  plot.caption = element_text(hjust = 0)
)
```

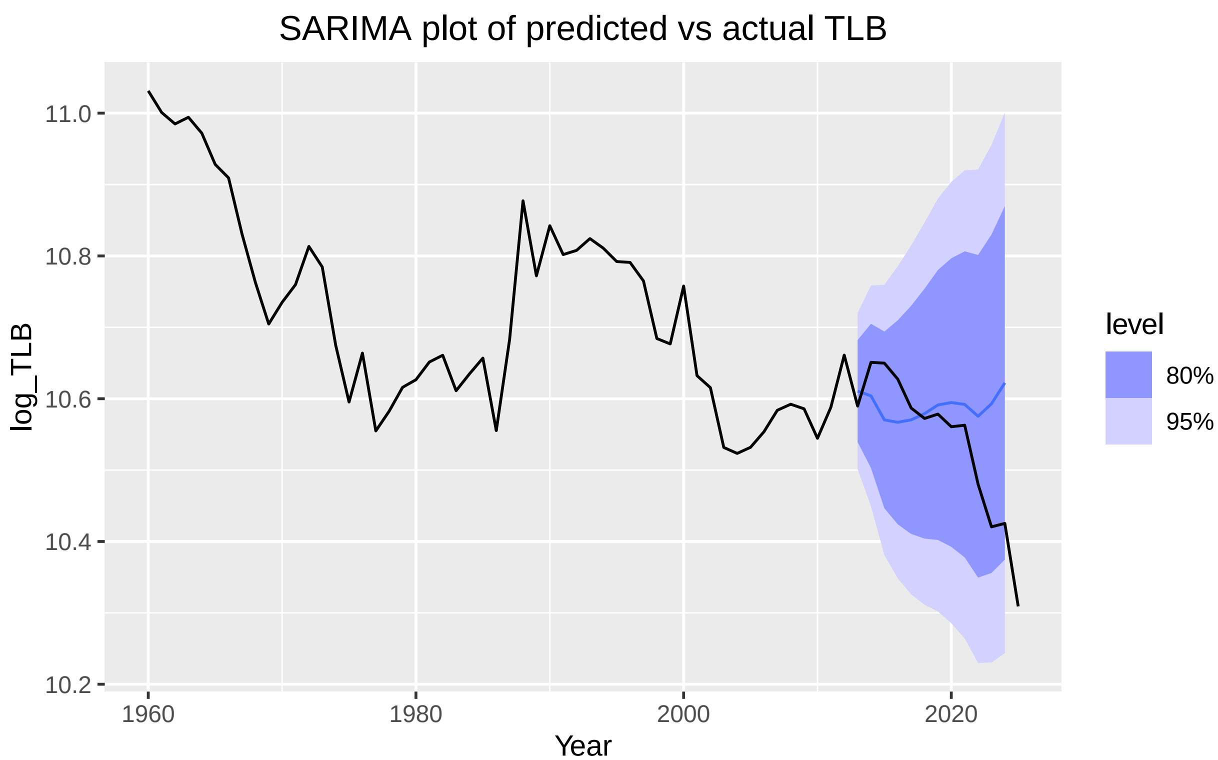


Figure 16. Plot shows that the SARIMA predictive model shows similar levels for the total live−births in singapore from 2013−2024 when compared to the known data.

Figure 16 shows that the predicted Total Live-Births from the SARIMA(1,1,0)(1,0,0)[12] model is on par with the known Total Live-Births data in Singapore from 2013-2024. Similarly to the Total Fertility-Rate figures 13 and 15, figure 16 and the ARIMA model seen in figure 14, also demonstrates how SARIMA is a better model to pick for this singapore data set. Similar to figure 15, we can see in figure 16 that the predictive model shows the data spiking at the end, whereas the actual data takes a sharp turn downwards.

```
tfr_resid <- augment(TFR_train_sarima)
```

```
tfr_resid |> ACF(.resid) |>
  autoplot() +
  labs(
    title = "ACF of SARIMA model for TFR ",
    caption = stringr::str_wrap("Figure 17. ACF Plot of the SARIMA model shows that there is a spike onl
  ) +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0)
  )
```

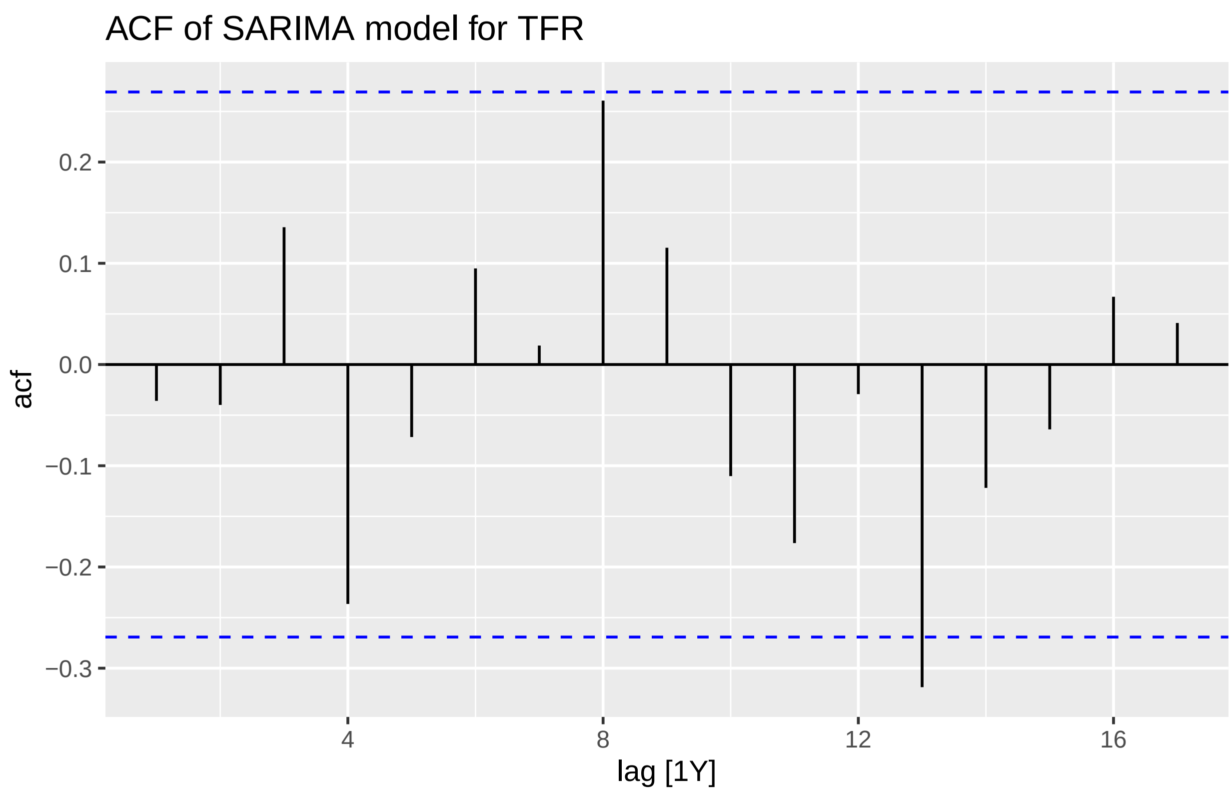


Figure 17. ACF Plot of the SARIMA model shows that there is a spike only at lag 13.

```
tfr_resid |> PACF(.resid) |>
  autoplot() +
  labs(
    title = "PACF of SARIMA model for TFR ",
    caption = stringr::str_wrap("Figure 18. PACF Plot of the SARIMA model shows that there is a spike a
  ) +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0)
  )
```

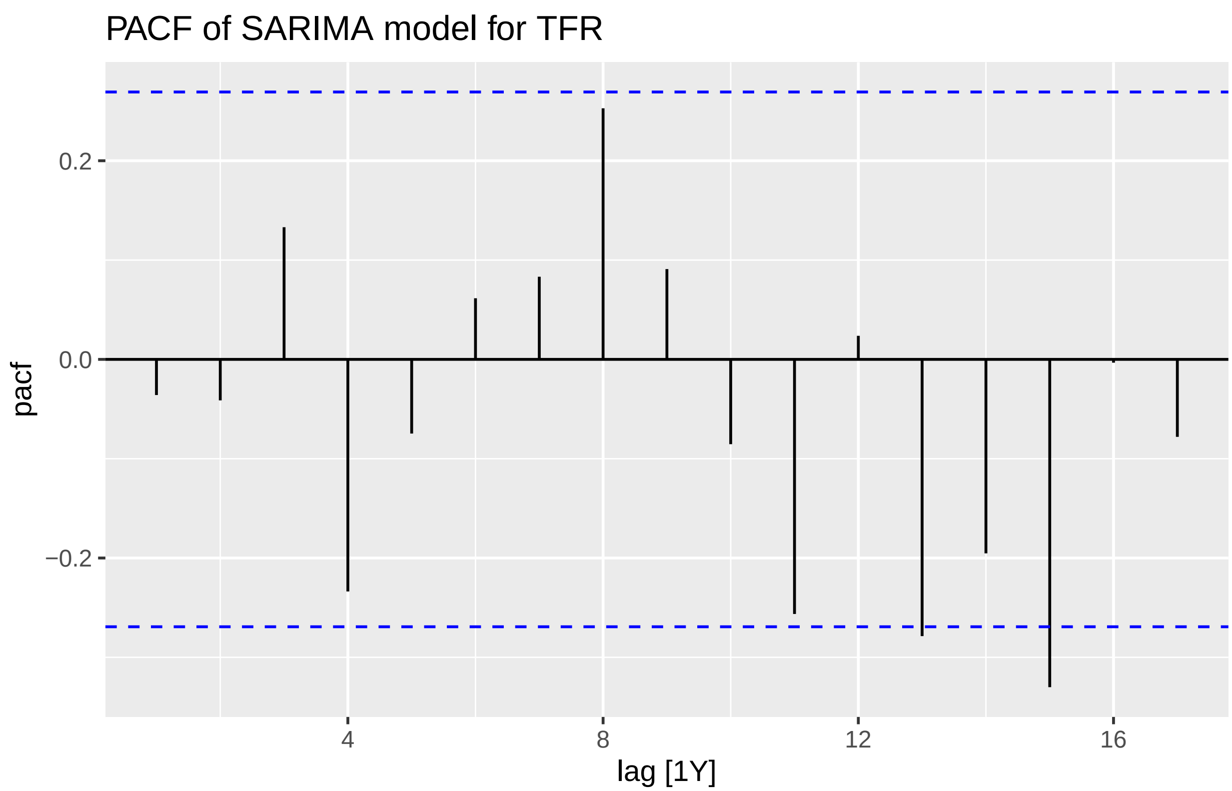


Figure 18. PACF Plot of the SARIMA model shows that there is a spike at lag 15 and potentially at lag 13.

Figures 17 and 18 show that some of the later lags spike, in figure 17 only lag 13 spikes, and in figure 18 lags 13 and 15 both spike. This shows that SARIMA model for the Total Fertility Rate in Singapore is fairly accurate.

```
tlb_resid <- augment(TLB_train_sarima)
```

```
tlb_resid |> ACF(.resid) |>
  autoplot()  +
  labs(
    title = "ACF of SARIMA model for TLB in Singapore ",
    caption = stringr::str_wrap("Figure 19. ACF Plot of the predictive SARIMA shows that there is a spil
  ) +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0)
  )
```

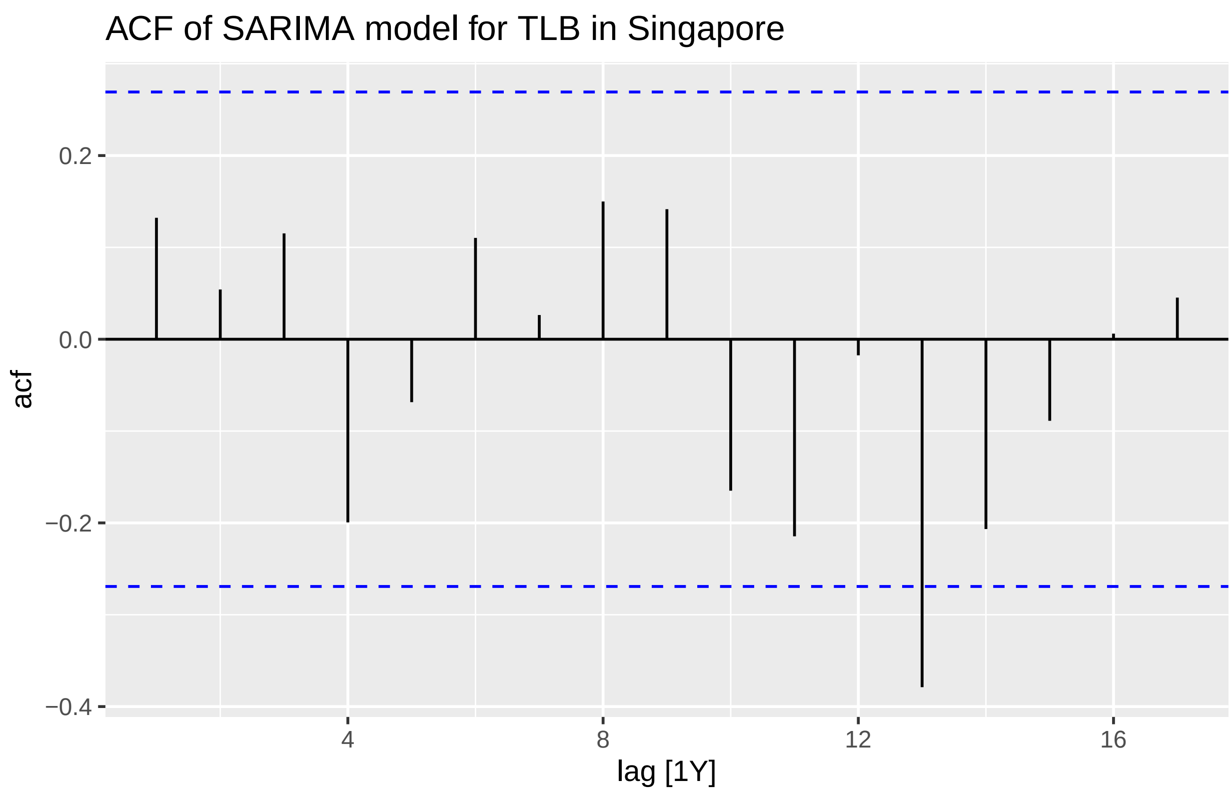


Figure 19. ACF Plot of the predictive SARIMA shows that there is a spike
only at lag 13.

```
tlb_resid |> PACF(.resid) |>
  autoplot()  +
  labs(
    title = "PACF of SARIMA model for TLB in Singapore ",
    caption = stringr::str_wrap("Figure 20. PACF Plot of the predictive SARIMA shows that there is a spi
  ) +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0)
  )
```

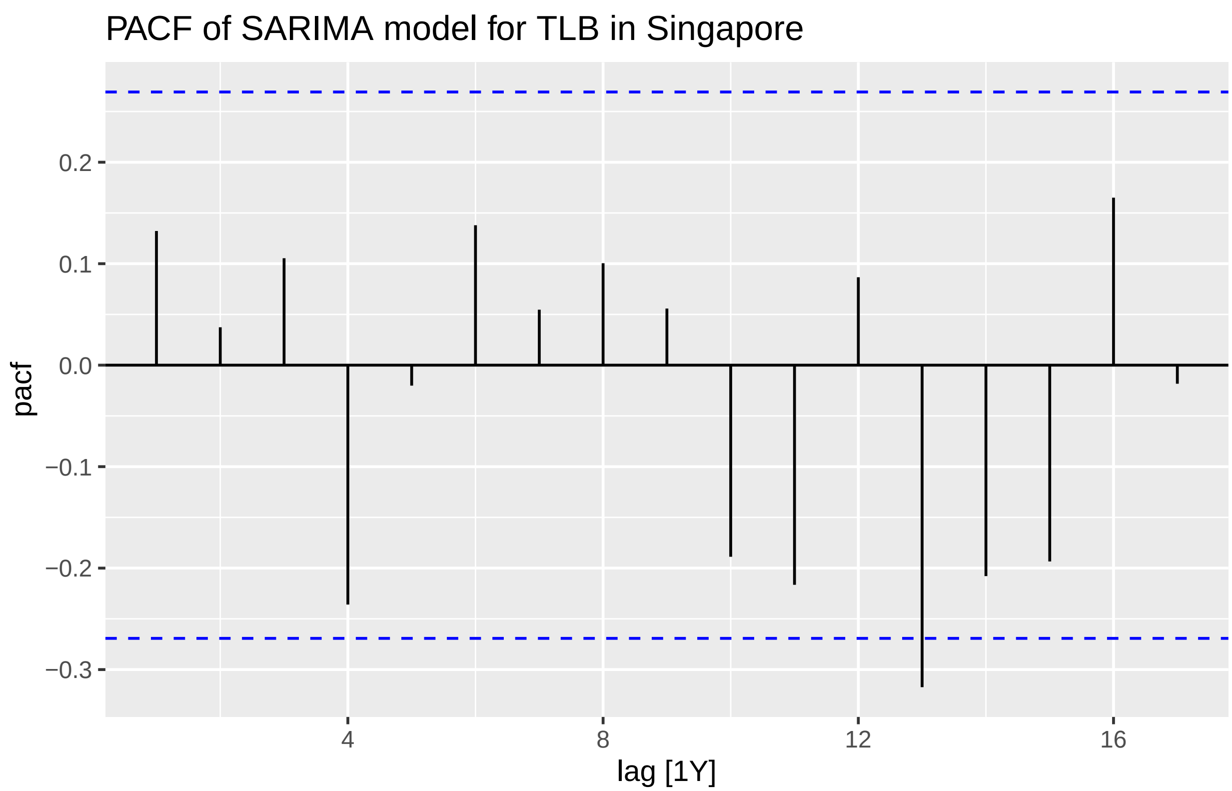


Figure 20. PACF Plot of the predictive SARIMA shows that there is a spike only at lag 13.

Figures 19 and 20 show the ACF and PACF for the Total Live-Births in Singapore. They both show spikes at lag 13.

Research Question: How can we statistically evaluate the Singaporean government family planning policies' effect on total live birth rate and total fertility rate?

The Singaporean government had family planning policies in place that pushed heavily for limiting your family size to two children or fewer. In 1966 a board was put together to deal with family planning in Singapore. They helped with educating on family planning (what it is and how to do it)and providing contraception. Sterilization was legalized in Singapore in 1970 and abortion was legalized in 1969. However, the government family planning policies specifically started in 1972 with the "Stop at Two" policy. This lasted until 1987 when the government shifted to a policy that encouraged couples to increase their family sizes.

```
singapore <- singapore |>
  mutate(
    small_family_push = (Year >=1972) & (Year <= 1986),
    big_family_push = (Year >=1987)
  )
```

```
singapore_train <- singapore |>
  filter(Year <= 2012)
```

```
TFR_policy_model <- singapore_train |>
  model(
    tfr_train_sarima = ARIMA(log_TFR ~ pdq() + PDQ(period = 12)),
    tfr_sarima_policy = ARIMA(log_TFR ~ small_family_push + big_family_push + pdq() + PDQ( period = 12)
```

```
  )
TFR_policy_model |> pivot_longer(everything(), names_to = "Model name", values_to = "Orders")
```

```
## # A mable: 2 x 2
## # Key:     Model name [2]
##   `Model name`                                             Orders
##   <chr>                                                   <model>
## 1 tfr_train_sarima                <ARIMA(1,2,1)(1,0,0)[12]>
## 2 tfr_sarima_policy <LM w/ ARIMA(1,0,3)(1,0,0)[12] errors>
```

Table 15. Shows that the chosen SARIMA model for the TFR training data (1960-2012) is SARIM(1,2,1)(1,0,0)[12], this is also seen in table 11. Table 15 also shows that the chosen SARIMA for the model that takes account for the policy years is LM with ARIMA(1,0,3)(1,0,0)[12].

```
glance(TFR_policy_model)
```

```
## # A tibble: 2 x 8
##   .model             sigma2 log_lik   AIC  AICc   BIC ar_roots   ma_roots
##   <chr>               <dbl>   <dbl> <dbl> <dbl> <dbl> <list>     <list>
## 1 tfr_train_sarima  0.00361    70.1 -132. -131. -125. <cpl [13]> <cpl [1]>
## 2 tfr_sarima_policy 0.00299    76.1 -136. -133. -120. <cpl [13]> <cpl [3]>
```

Table 16. Shows that the σ^2, AIC, and BIC were lower for the model that take the government policies into account.

It can be seen from table 16 that the model that includes the government policies is a better model than the regular sarima model as it has a lower σ^2, AIC, and BIC.

```
singapore_future <- singapore |>
  filter(Year >=2013)
```

```
TFR_policy_fc <- TFR_policy_model |>
  select(tfr_sarima_policy) |>
  forecast(new_data = singapore_future)
```

```
TFR_policy_fc |>
  autoplot() +
  autolayer(singapore, log_TFR) +
  labs(
    title = "Predictive Model with Policies vs Know TFR",
    caption = stringr::str_wrap("Figure 21. Plot of the predictive SARIMA shows that the SARIMA model t
  ) +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0)
  )
```

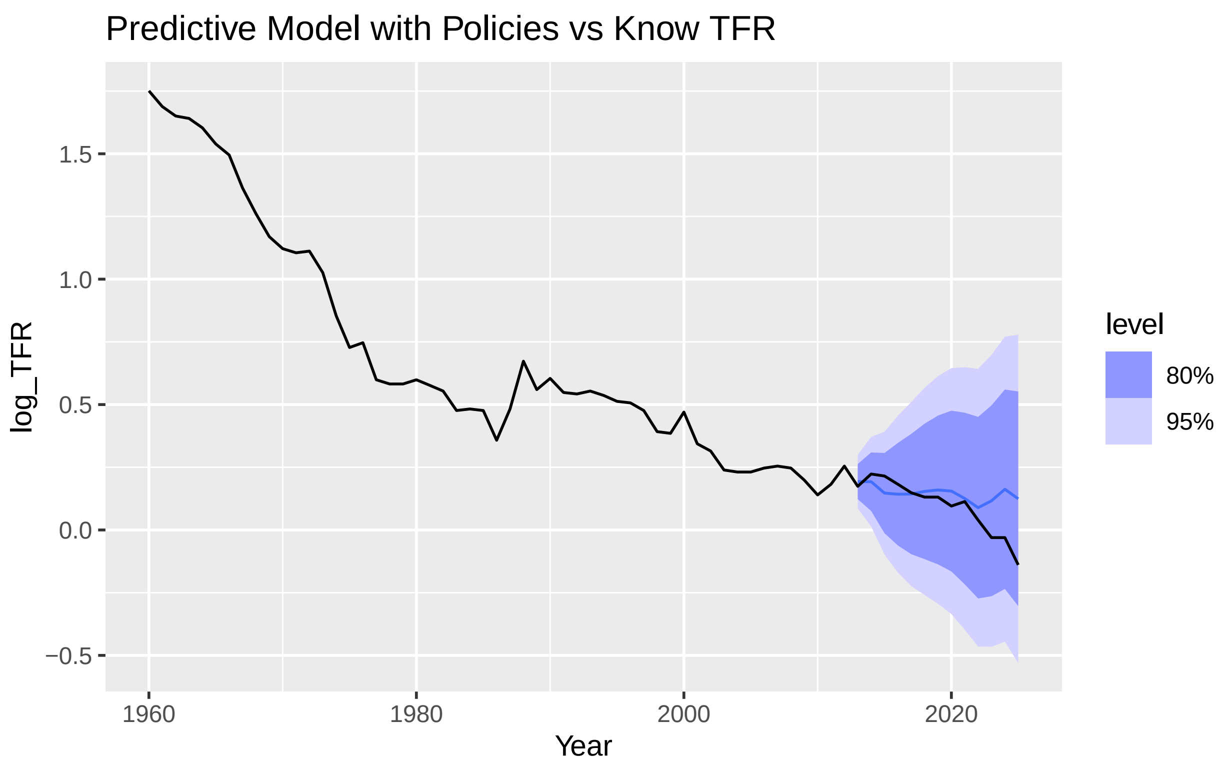


Figure 21. Plot of the predictive SARIMA shows that the SARIMA model that takes the government's family planning policies into account, follows the known TFR data fairly well.

Both figure 21 and table 16 confirm that the predictive models that account for the two different stages of Singaporean government family planning policies is better at predicting the values for the Total Fertility Rate in Singapore from 2013-2024 than a SARIMA model that does not take these policies into consideration. Figure 15 shows the plot of the SARIMA plot predicting the values for the 2013-2024 period compared to the actual data, and it was seen that at the end of the time period the model predicts the TFR will increase, whereas the actual data downturns. In figure 21, it can be seen that while the predictive data does not drop as much as the actual data does, it is downturned, like the actual data.

```
TLB_policy_model <- singapore_train |>
  model(
    tlb_train_sarima = ARIMA(log_TLB ~ pdq() + PDQ(period = 12)),
    tlb_sarima_policy = ARIMA(log_TLB ~ small_family_push + big_family_push + pdq() + PDQ( period = 12)
  )
TLB_policy_model |> pivot_longer(everything(), names_to = "Model name", values_to = "Orders")
```

```
## # A mable: 2 x 2
## # Key:     Model name [2]
##   `Model name`                                     Orders
##   <chr>                                           <model>
## 1 tlb_train_sarima                <ARIMA(0,1,0)(1,0,0)[12]>
## 2 tlb_sarima_policy <LM w/ ARIMA(1,0,0)(1,0,0)[12] errors>
```

Table 17. Shows that the chosen SARIMA model for the TLB training data (1960-2012) is SARIMA(0,1,0)(1,0,0)[12], this is also seen in table 14. Table 15 also shows that the chosen SARIMA for the model that takes account for the policy years is LM with ARIMA(1,0,0)(1,0,0)[12].

```r
glance(TLB_policy_model)
```

```
## # A tibble: 2 x 8
##   .model             sigma2 log_lik   AIC  AICc   BIC ar_roots   ma_roots
##   <chr>               <dbl>   <dbl> <dbl> <dbl> <dbl> <list>     <list>
## 1 tlb_train_sarima  0.00311    75.8 -148. -147. -144. <cpl [12]> <cpl [0]>
## 2 tlb_sarima_policy 0.00254    82.1 -152. -150. -140. <cpl [13]> <cpl [0]>
```

Table 18. Shows that the σ^2, AIC, and BIC were lower for the model that take the government policies into account.

It can be seen from table 18 that the model that includes the government policies for the Total Live-Births is a better model than the regular sarima model as it has a lower σ^2, AIC, and BIC.

```r
TLB_policy_fc <- TLB_policy_model |>
  select(tlb_sarima_policy) |>
  forecast(new_data = singapore_future)
```

```r
TLB_policy_fc |>
  autoplot() +
  autolayer(singapore, log_TLB) +
  labs(
    title = "Predictive Model with Policies vs Know TLB",
    caption = stringr::str_wrap("Figure 22. Plot of the predictive SARIMA shows that the SARIMA model t
  ) +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0)
  )
```

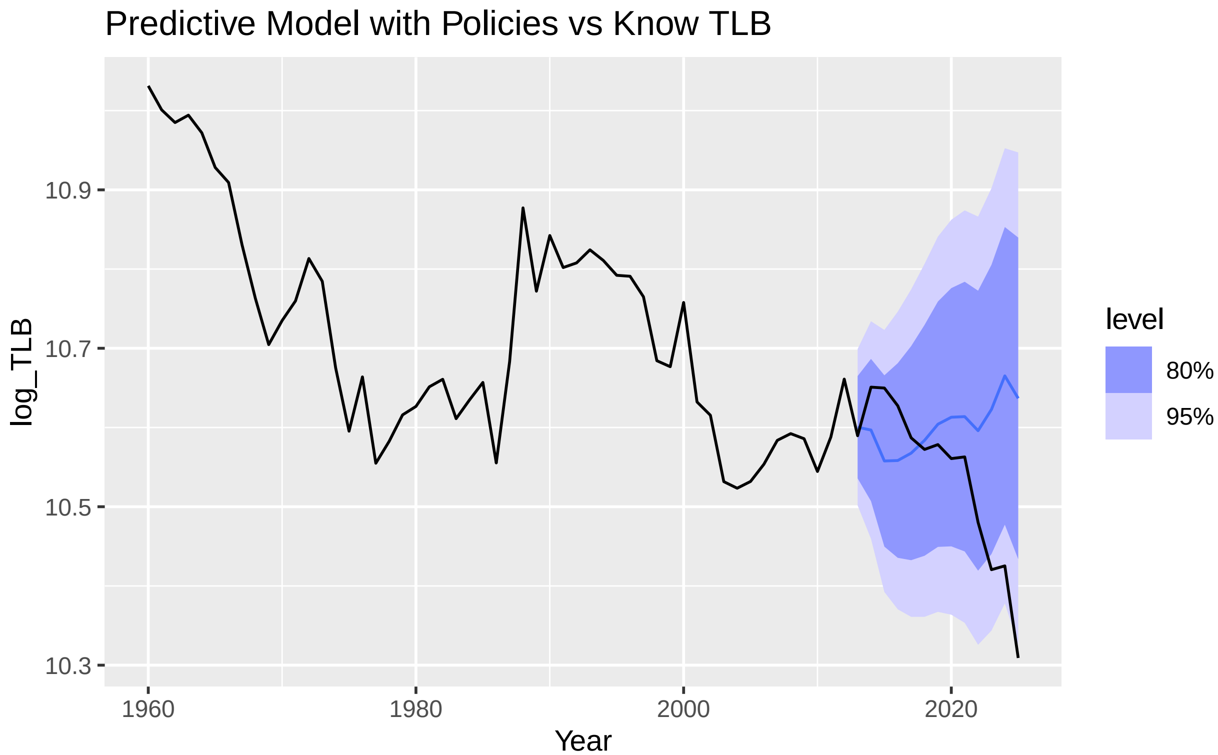
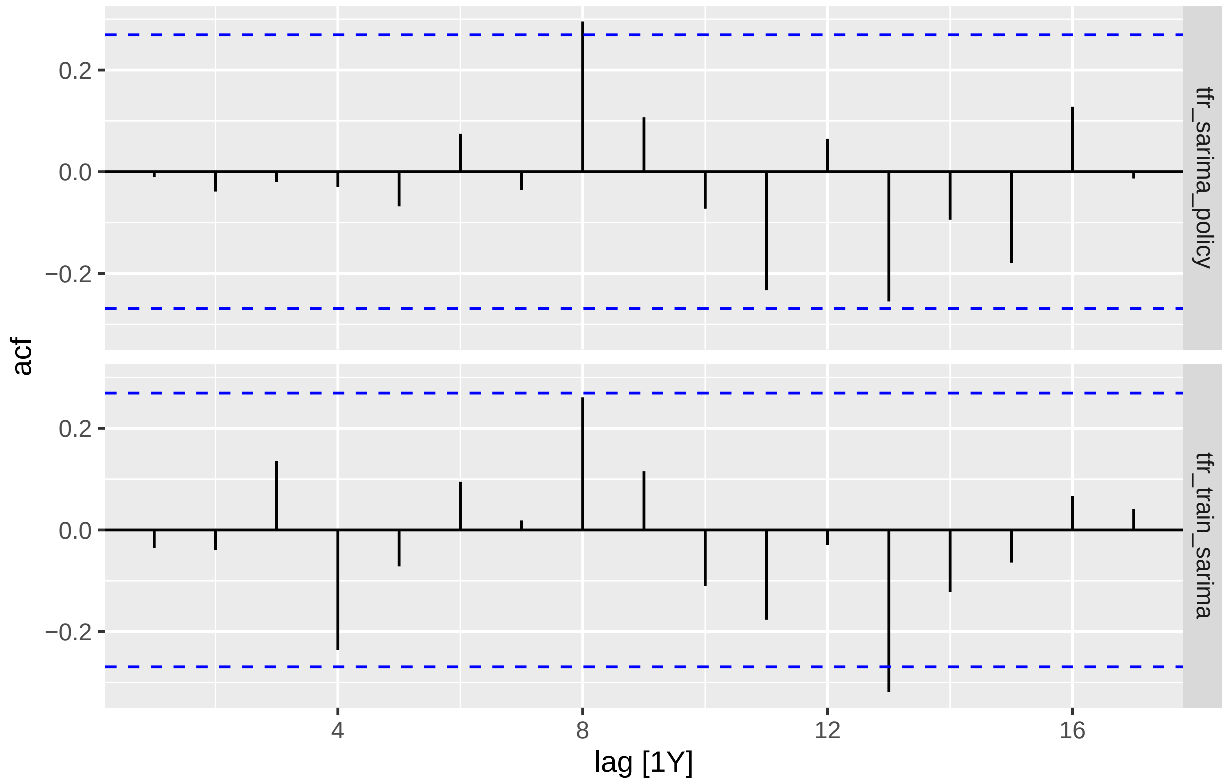


Figure 22. Plot of the predictive SARIMA shows that the SARIMA model that takes the government's family planning policies into account, follows the known TLB data fairly well.

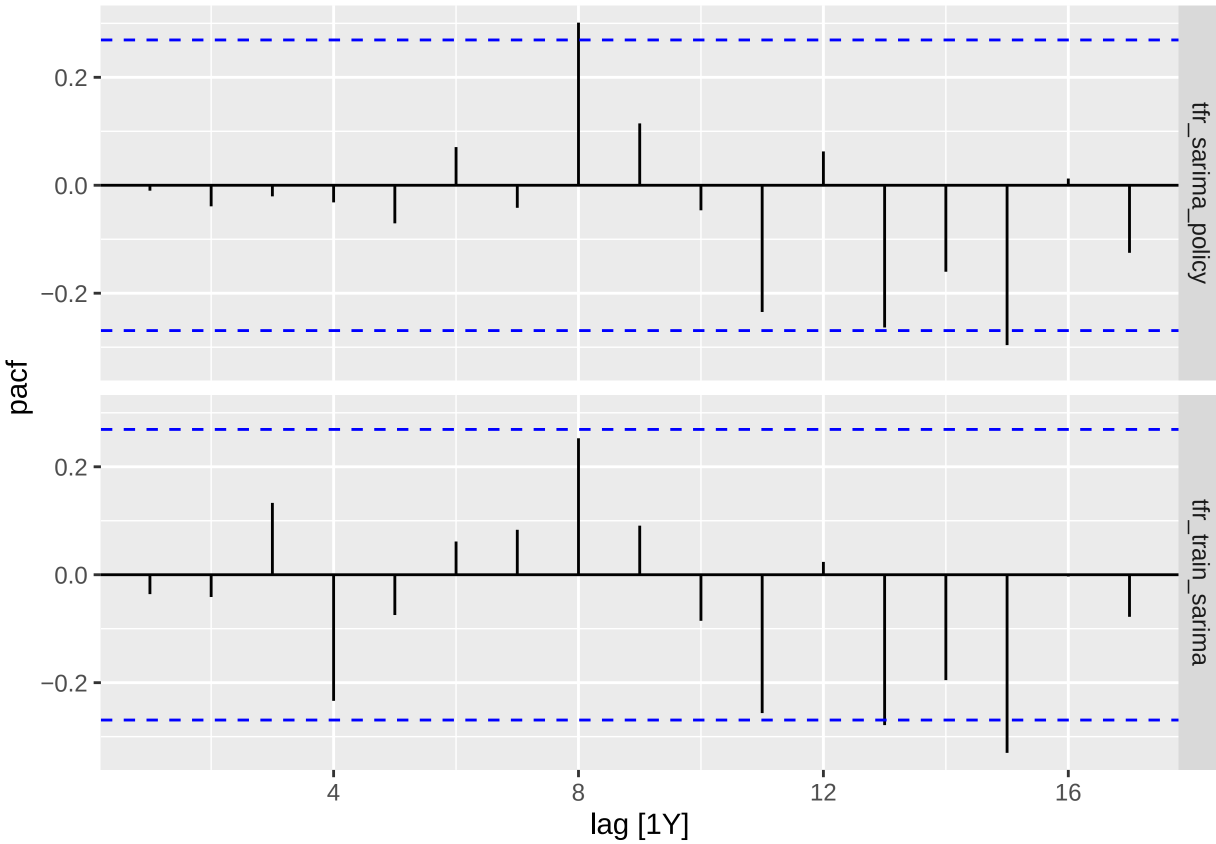
It can be seen that figure 22 follows the actual data fairly well, but it does not drop nearly as much as the actual data does around 2021.

```
tfr_policy_resid <- augment(TFR_policy_model)
tfr_policy_resid |>
  ACF(.resid) |>
  autoplot() +
  labs( caption = stringr::str_wrap("tfr_sarima_policy = ARIMA(log_TFR ~ small_family_push + big_family_
  theme(plot.caption = element_text(hjust = 0)
  )
```

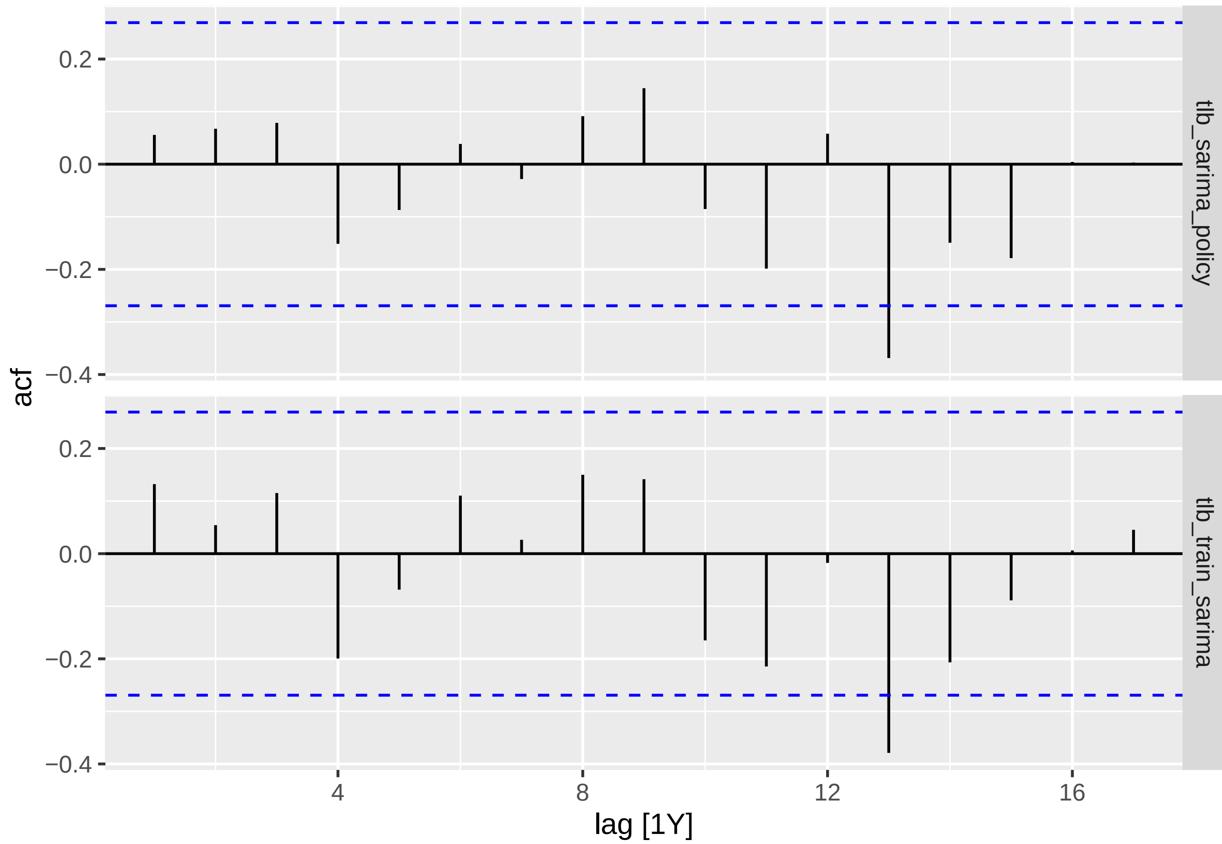


tfr_sarima_policy = ARIMA(log_TFR ~ small_family_push + big_family_push + pdq()
+ PDQ(period = 12))

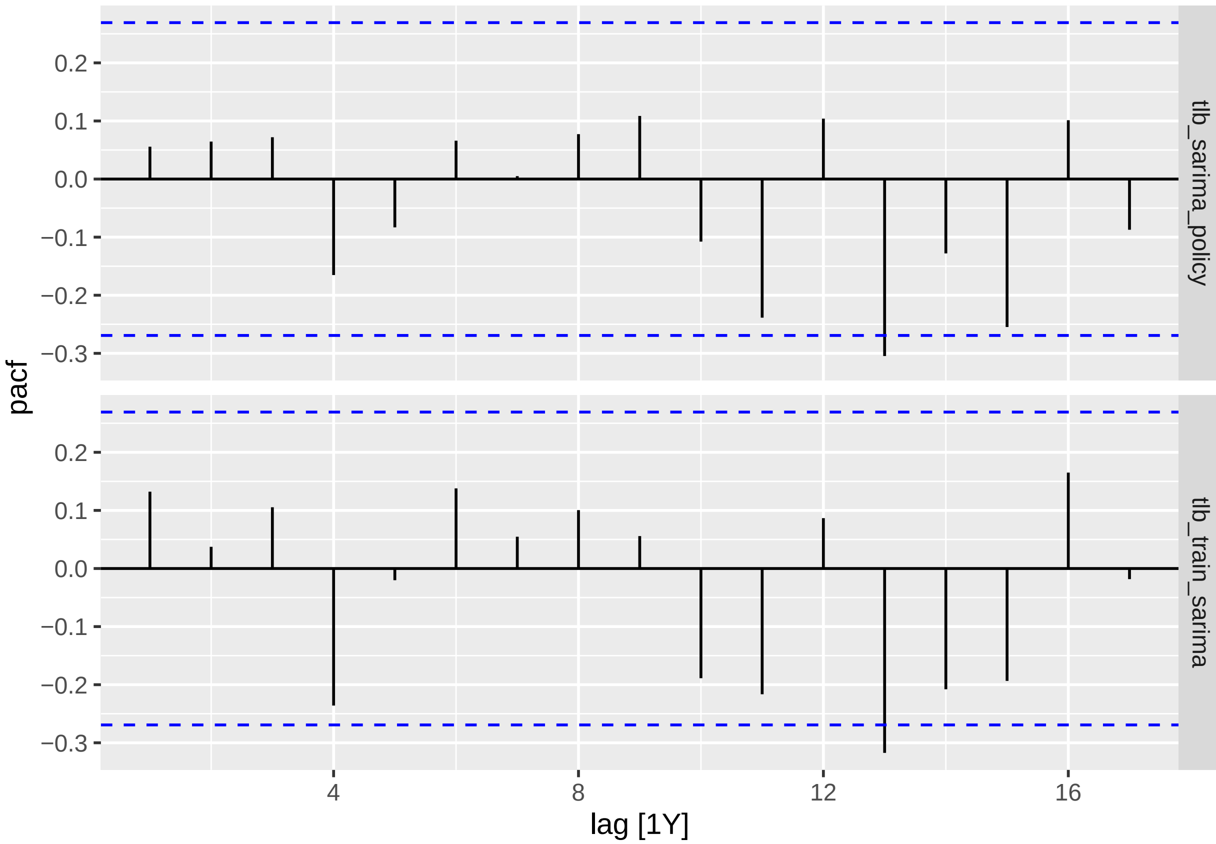
```
tfr_policy_resid |> PACF(.resid) |>
  autoplot()
```




```
tlb_policy_resid <- augment(TLB_policy_model)
tlb_policy_resid |> ACF(.resid) |>
  autoplot()
```



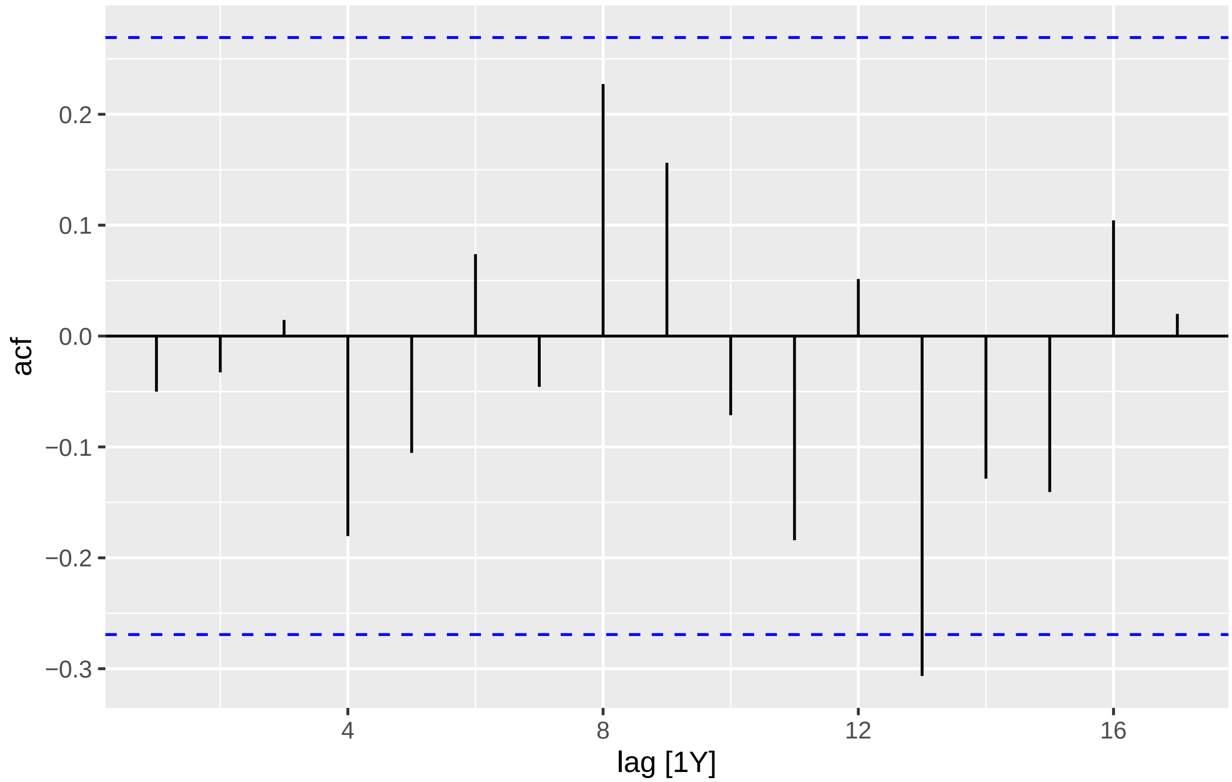

```
tlb_policy_resid |> PACF(.resid) |>
  autoplot()
```




```
TFR_perfect_1 <-singapore_train |>
  model(
    tfr_sarima_policy_1 = ARIMA(log_TFR ~ small_family_push + big_family_push + pdq(2,0,2) + PDQ(period
  )
TFR_perfect_1 |> pivot_longer(everything(), names_to = "Model name", values_to = "Orders")
```

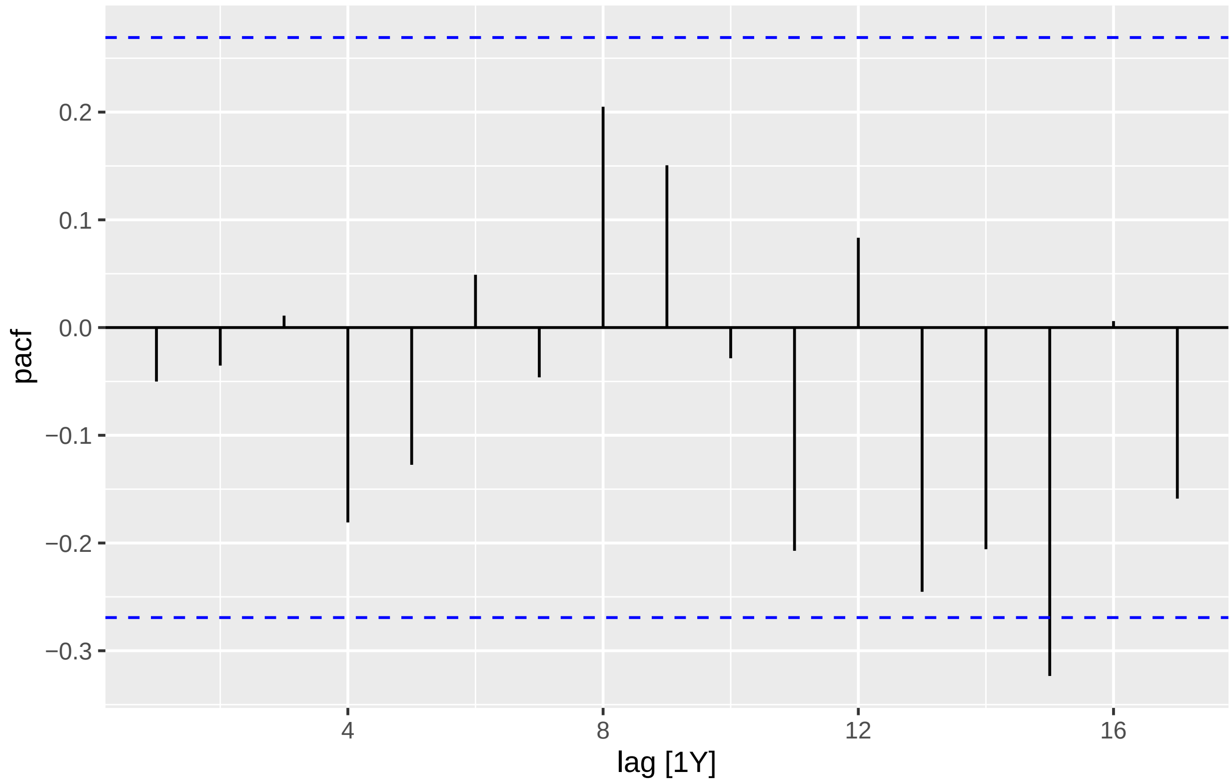
```
## # A mable: 1 x 2
## # Key:     Model name [1]
##   `Model name`                                           Orders
##   <chr>                                                 <model>
## 1 tfr_sarima_policy_1 <LM w/ ARIMA(2,0,2)(1,0,1)[12] errors>
```

```
tfr_perfect1_resid <- augment(TFR_perfect_1)
tfr_perfect1_resid |>
  ACF(.resid) |>
  autoplot() +
  labs( caption = stringr::str_wrap("tfr_sarima_policy_1 = ARIMA(log_TFR ~ small_family_push + big_famil
  theme(plot.caption = element_text(hjust = 0)
  )
```



tfr_sarima_policy_1 = ARIMA(log_TFR ~ small_family_push + big_family_push + pdq(2,0,2) + PDQ(period = 12))

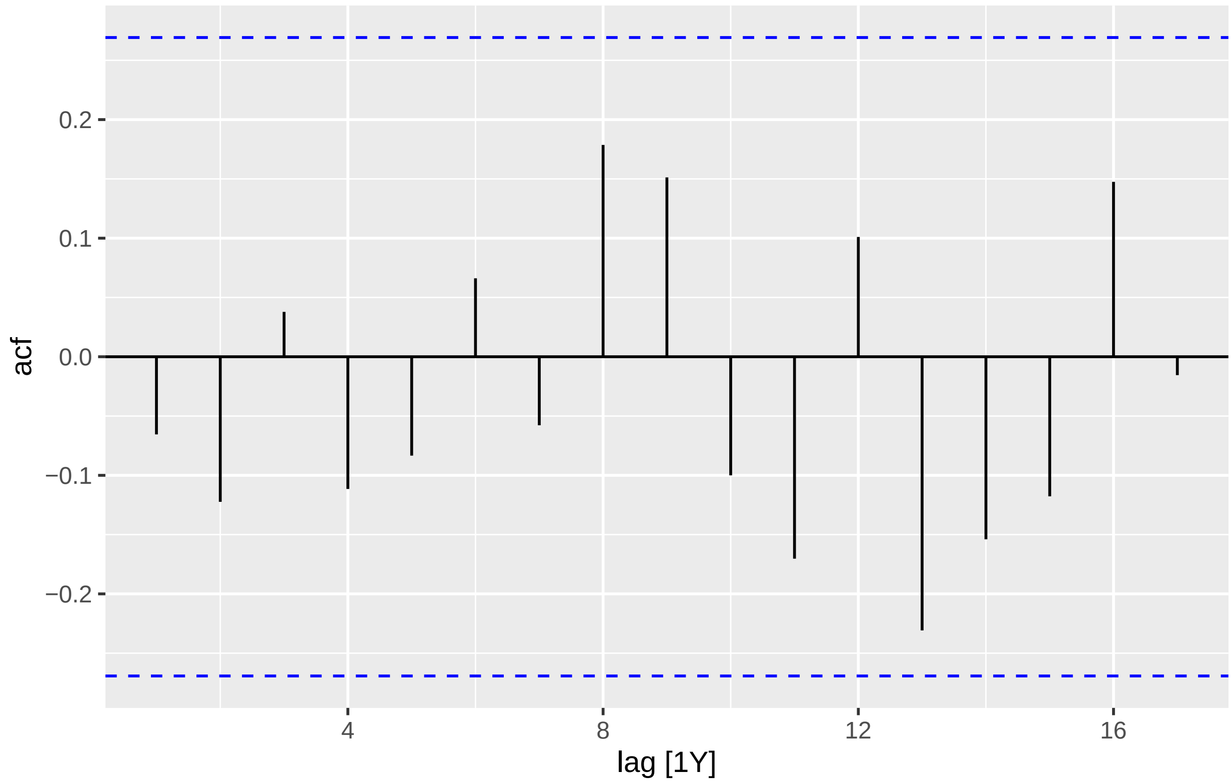
```
tfr_perfect1_resid |>
  PACF(.resid) |>
  autoplot() +
  labs( caption = stringr::str_wrap("tfr_sarima_policy_1 = ARIMA(log_TFR ~ small_family_push + big_famil
  theme(plot.caption = element_text(hjust = 0)
  )
```



tfr_sarima_policy_1 = ARIMA(log_TFR ~ small_family_push + big_family_push +
pdq(2,0,2) + PDQ(period = 12))

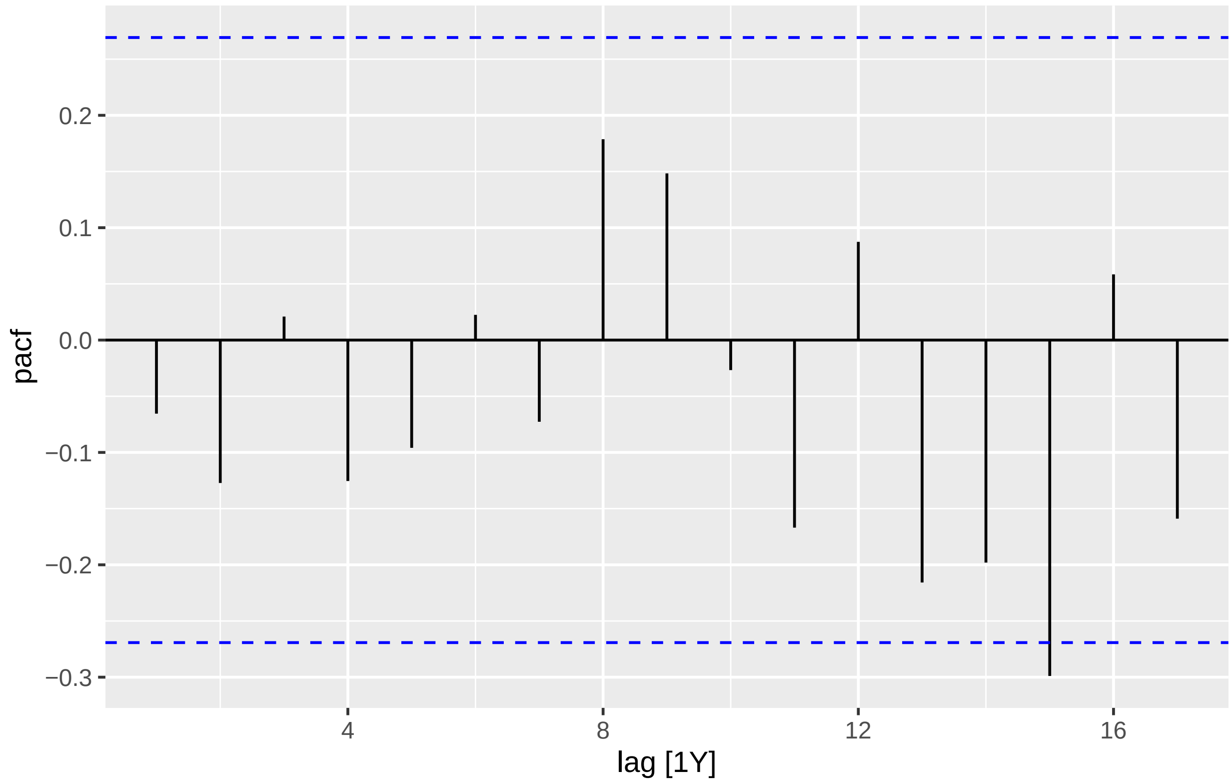
```
TFR_perfect_2 <-singapore_train |>
  model(
    tfr_sarima_policy_2 = ARIMA(log_TFR ~ small_family_push + big_family_push + pdq(2,0,2) + PDQ(1,0,0,
  )
```

```
tfr_perfect2_resid <- augment(TFR_perfect_2)
tfr_perfect2_resid |>
  ACF(.resid) |>
  autoplot() +
  labs( caption = stringr::str_wrap("tfr_sarima_policy_2 = ARIMA(log_TFR ~ small_family_push + big_famil
  theme(plot.caption = element_text(hjust = 0)
  )
```



tfr_sarima_policy_2 = ARIMA(log_TFR ~ small_family_push + big_family_push +
pdq(2,0,2) + PDQ(1,0,0, period = 12))

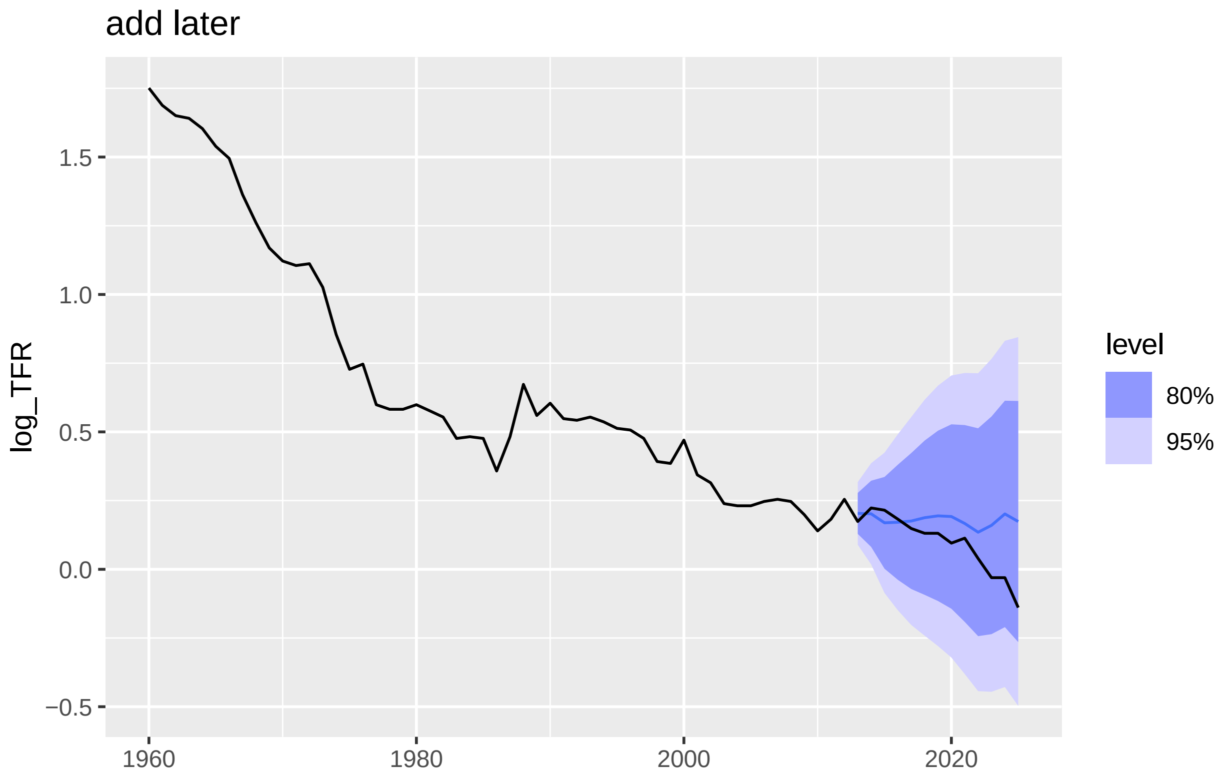
```
tfr_perfect2_resid |>
  PACF(.resid) |>
  autoplot() +
  labs( caption = stringr::str_wrap("tfr_sarima_policy_2 = ARIMA(log_TFR ~ small_family_push + big_famil
  theme(plot.caption = element_text(hjust = 0))
  )
```



tfr_sarima_policy_2 = ARIMA(log_TFR ~ small_family_push + big_family_push +
pdq(2,0,2) + PDQ(1,0,0, period = 12))

```
TFR_perfect_2_fc <- TFR_perfect_2 |>
  forecast(new_data = singapore_future)
```

```
TFR_perfect_2_fc |>
  autoplot() +
  autolayer(singapore, log_TFR) +
  labs(
    title = "add later",
    caption = stringr::str_wrap("add later", width = 75)
  ) +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0)
  )
```

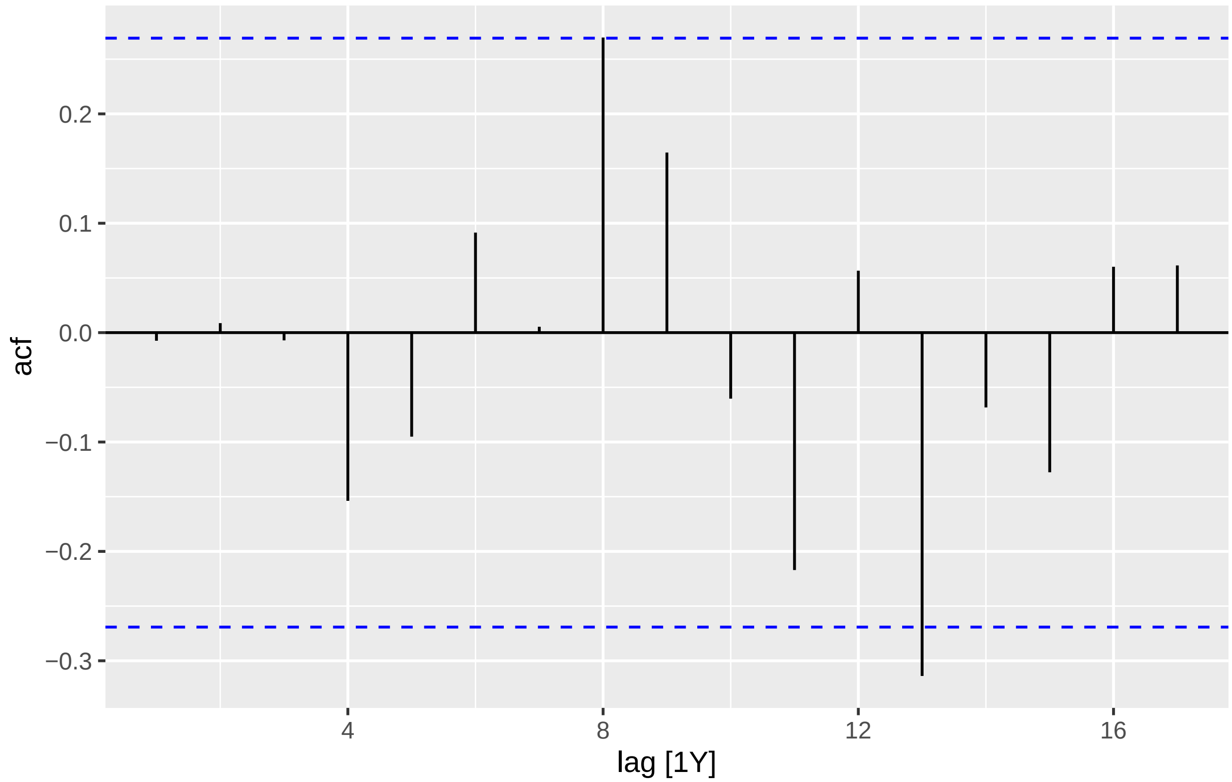


add later

```
TFR_perfect_3 <-singapore_train |>
  model(
    tfr_sarima_policy_3 = ARIMA(log_TFR ~ small_family_push + big_family_push + pdq(2,1,2) + PDQ( perioc
  )
TFR_perfect_3 |> pivot_longer(everything(), names_to = "Model name", values_to = "Orders")
```

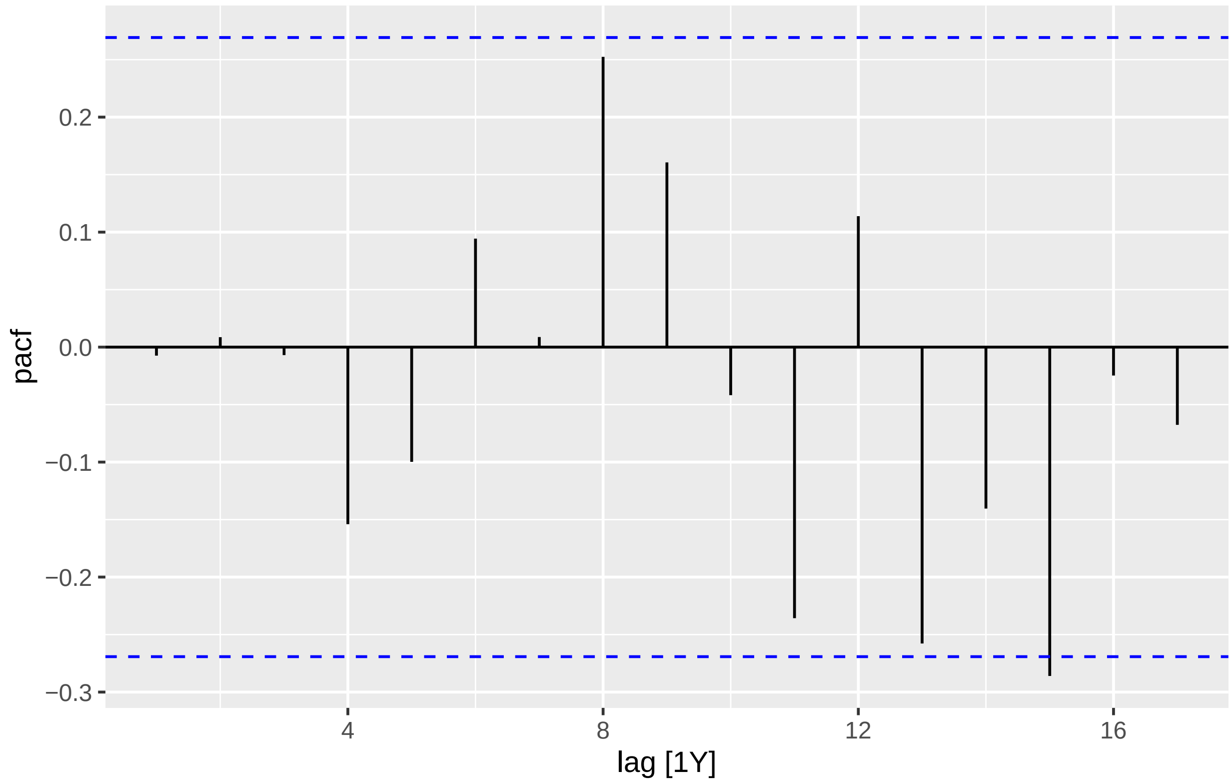
```
## # A mable: 1 x 2
## # Key:     Model name [1]
##   `Model name`                                              Orders
##   <chr>                                                    <model>
## 1 tfr_sarima_policy_3 <LM w/ ARIMA(2,1,2)(1,0,0)[12] errors>
```

```
tfr_perfect3_resid <- augment(TFR_perfect_3)
tfr_perfect3_resid |>
  ACF(.resid) |>
  autoplot() +
  labs( caption = stringr::str_wrap("tfr_sarima_policy_3 = ARIMA(log_TFR ~ small_family_push + big_famil
  theme(plot.caption = element_text(hjust = 0)
  )
```



tfr_sarima_policy_3 = ARIMA(log_TFR ~ small_family_push + big_family_push +
pdq(2,1,2) + PDQ(period = 12))

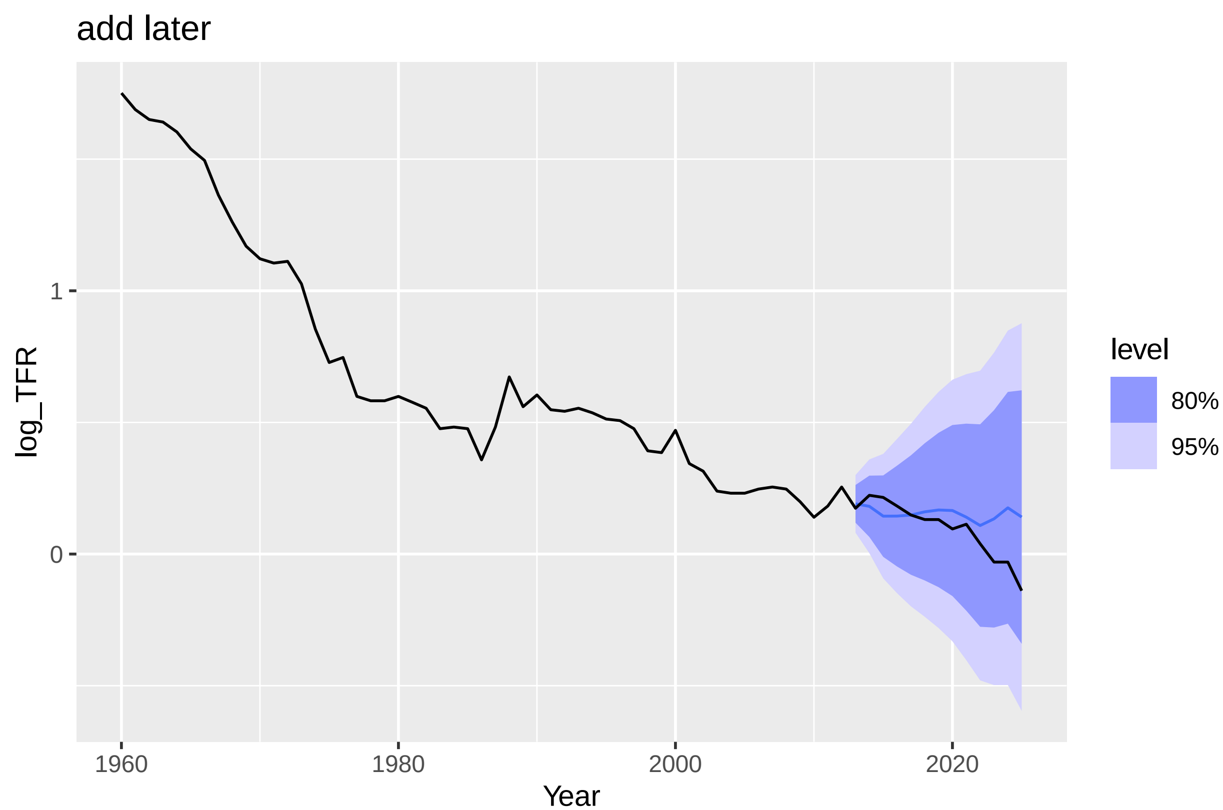
```
tfr_perfect3_resid |>
  PACF(.resid) |>
  autoplot() +
  labs( caption = stringr::str_wrap("tfr_sarima_policy_3 = ARIMA(log_TFR ~ small_family_push + big_famil
  theme(plot.caption = element_text(hjust = 0)
  )
```



tfr_sarima_policy_3 = ARIMA(log_TFR ~ small_family_push + big_family_push + pdq(2,1,2) + PDQ(period = 12))

```
TFR_perfect_3_fc <- TFR_perfect_3 |>
  forecast(new_data = singapore_future)
```

```
TFR_perfect_3_fc |>
  autoplot() +
  autolayer(singapore, log_TFR) +
  labs(
    title = "add later",
    caption = stringr::str_wrap("add later", width = 75)
  ) +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0)
  )
```



add later

```
TFR_no_policy <- singapore_train |>
  model(
    tfr_nopolicy_sarima = ARIMA(log_TFR ~ pdq(2,0,2) + PDQ(period = 12))
  )
```

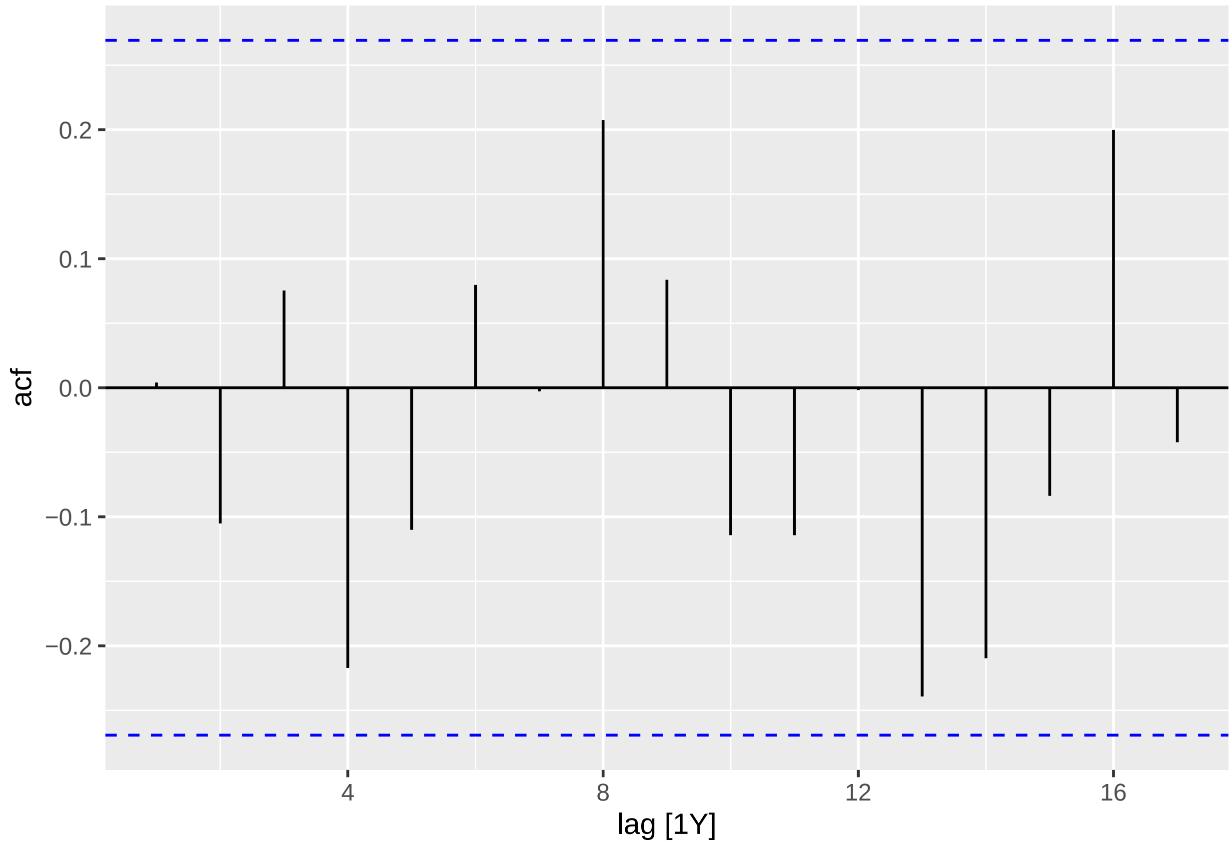
```
## Warning in sqrt(diag(best$var.coef)): NaNs produced
```

```
TFR_no_policy |> pivot_longer(everything(), names_to = "Model name", values_to = "Orders")
```

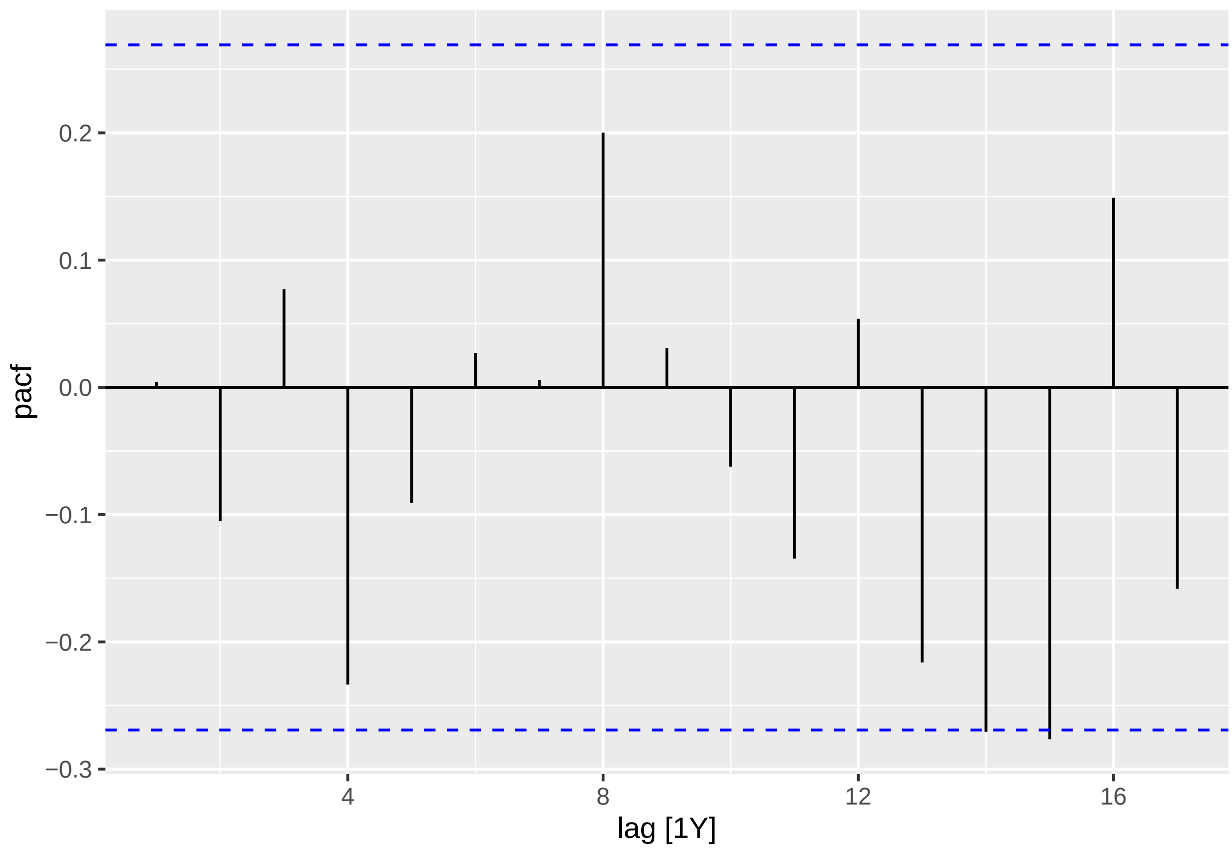
```
## # A mable: 1 x 2
## # Key:     Model name [1]
##   `Model name`                                       Orders
##   <chr>                                             <model>
## 1 tfr_nopolicy_sarima <ARIMA(2,0,2)(1,0,0)[12] w/ mean>
```

```
tfr_no_policy_resid <- augment(TFR_no_policy)

# ACF Plot
tfr_no_policy_resid |>
  ACF(.resid) |>
  autoplot()
```

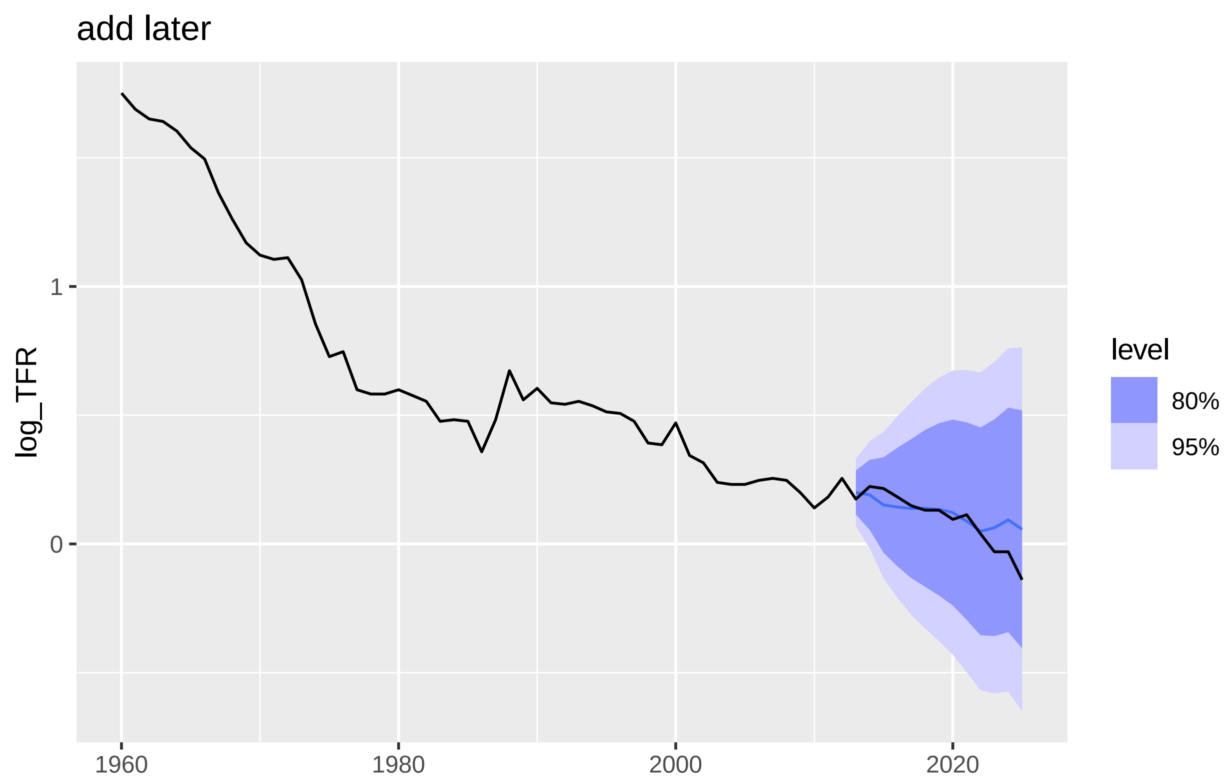



```r
# PACF Plot
tfr_no_policy_resid |>
  PACF(.resid) |>
  autoplot()
```




```r
TFR_no_policy_fc <- TFR_no_policy |>
  forecast(new_data = singapore_future)
```

```r
TFR_no_policy_fc |>
  autoplot() +
  autolayer(singapore, log_TFR) +
  labs(
    title = "add later",
    caption = stringr::str_wrap("add later", width = 75)
  ) +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0)
  )
```

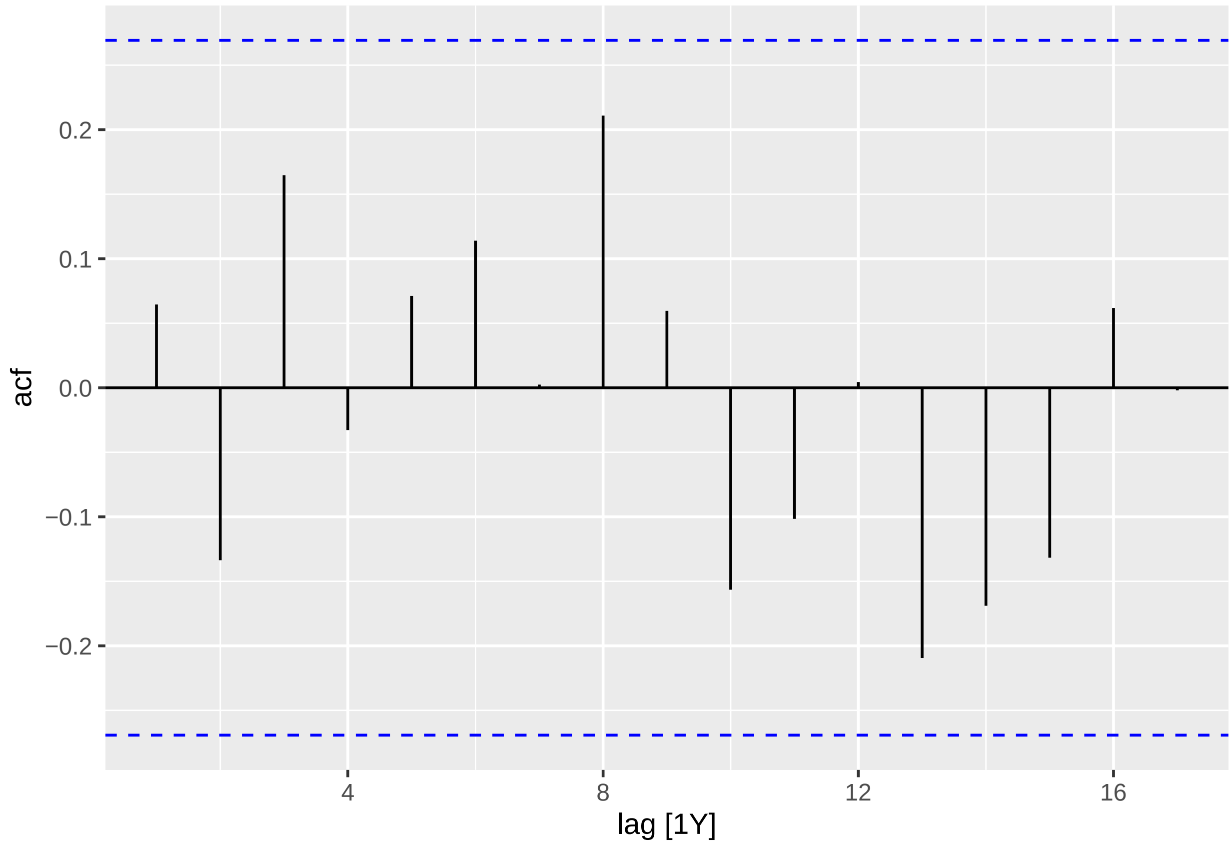


add later

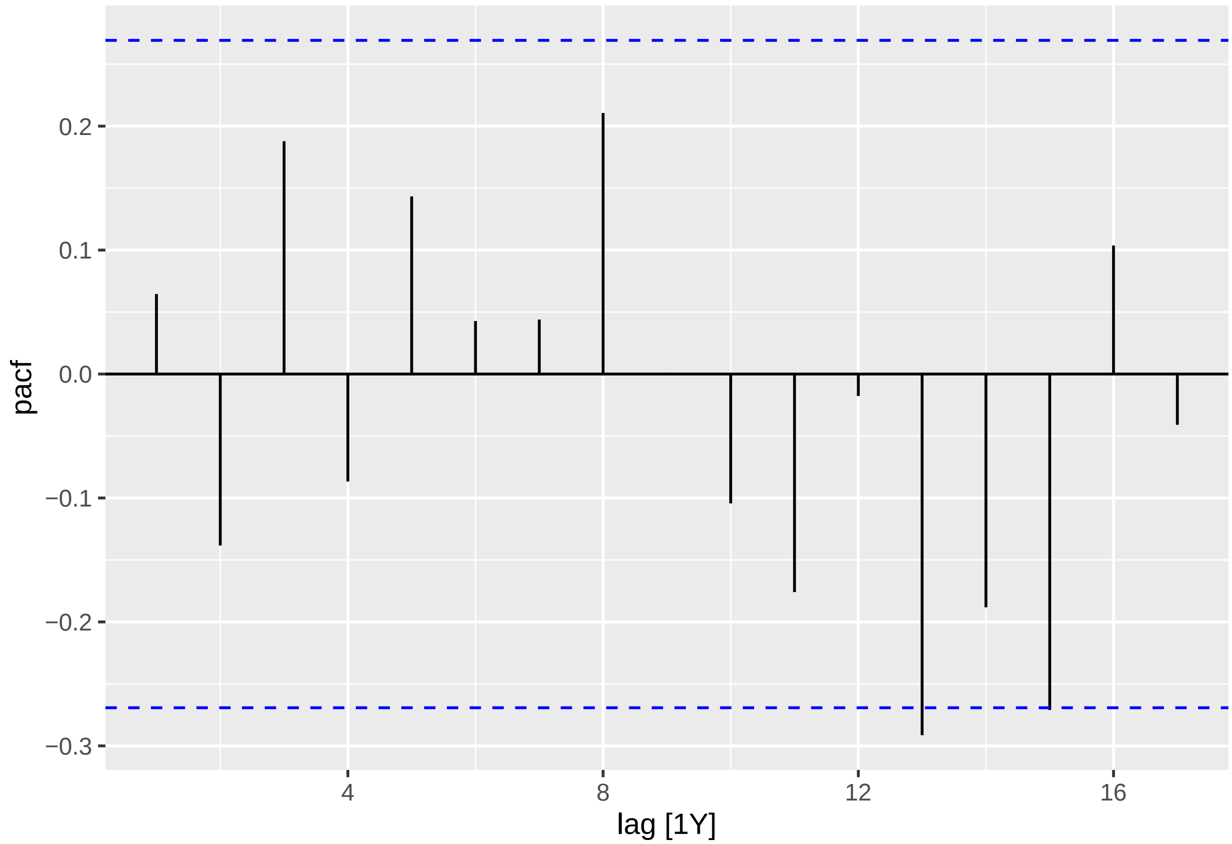
```
TFR_no_policy2 <- singapore_train |>
  model(
    tfr_base_sarima2 = ARIMA(log_TFR ~ pdq(2,1,2) + PDQ(period = 12))
  )

tfr_no_policy_resid2 <- augment(TFR_no_policy2)

# ACF Plot
tfr_no_policy_resid2 |>
  ACF(.resid) |>
  autoplot()
```

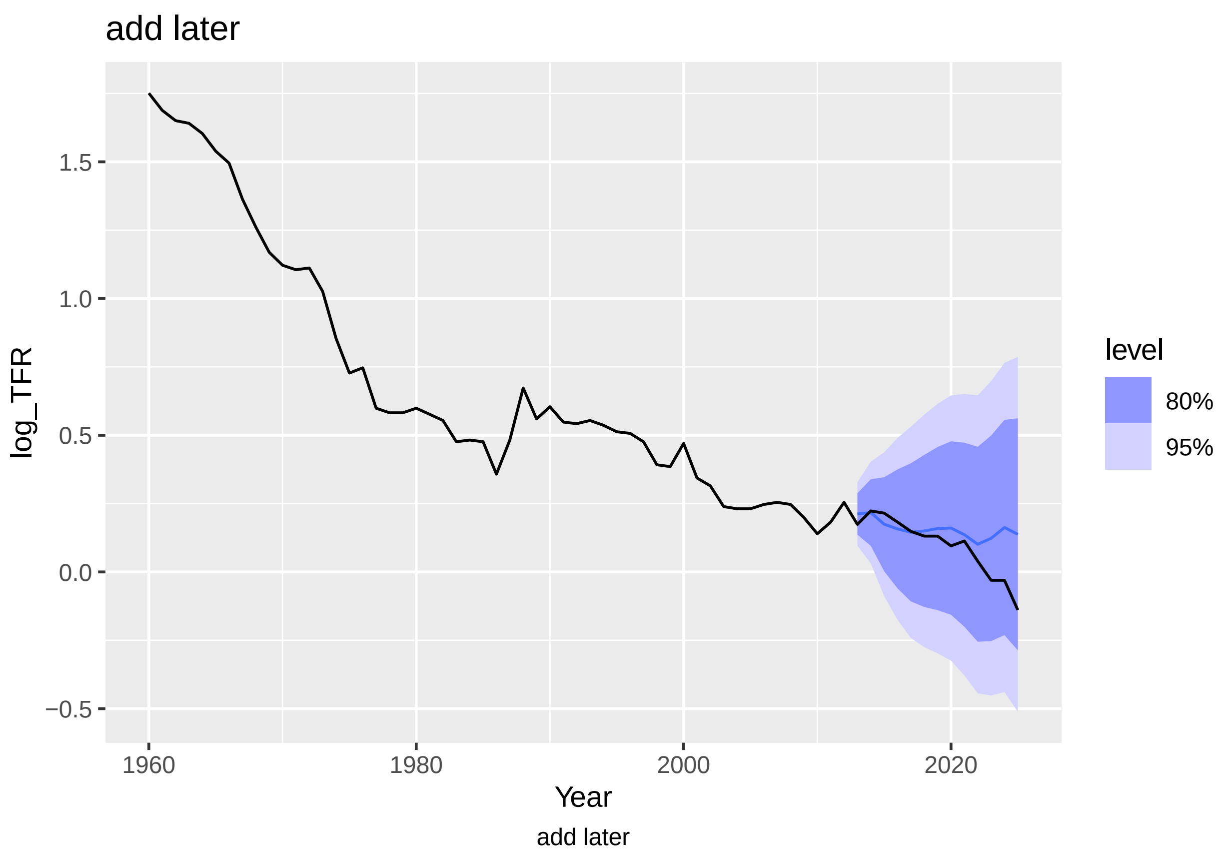



```r
# PACF Plot
tfr_no_policy_resid2 |>
  PACF(.resid) |>
  autoplot()
```




```r
TFR_no_policy2_fc <- TFR_no_policy2 |>
  forecast(new_data = singapore_future)
```

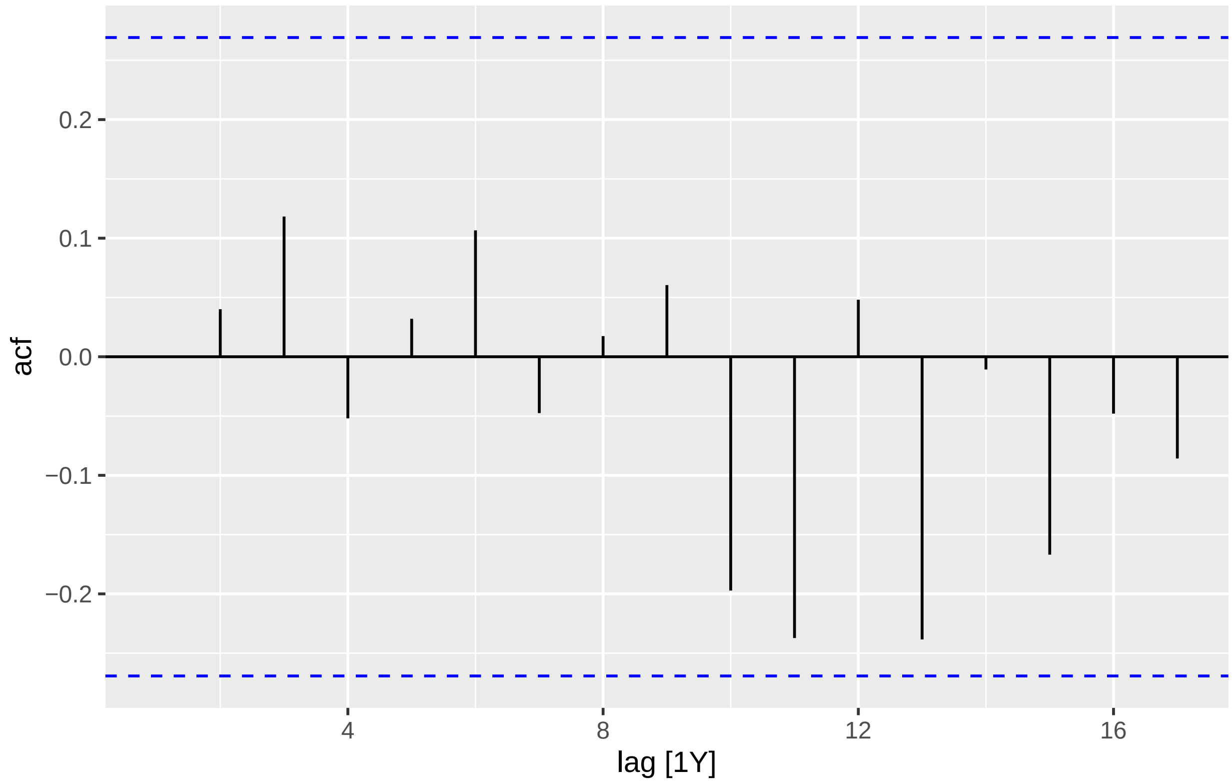
```r
TFR_no_policy2_fc |>
  autoplot() +
  autolayer(singapore, log_TFR) +
  labs(
    title = "add later",
    caption = stringr::str_wrap("add later", width = 75)
  ) +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0)
  )
```




```
TLB_perfect <-singapore_train |>
  model(
    tlb_sarima_policy = ARIMA(log_TLB ~ small_family_push + big_family_push + pdq(2,1,2) + PDQ(1,0,1, p
  )
TLB_perfect |> pivot_longer(everything(), names_to = "Model name", values_to = "Orders")
```

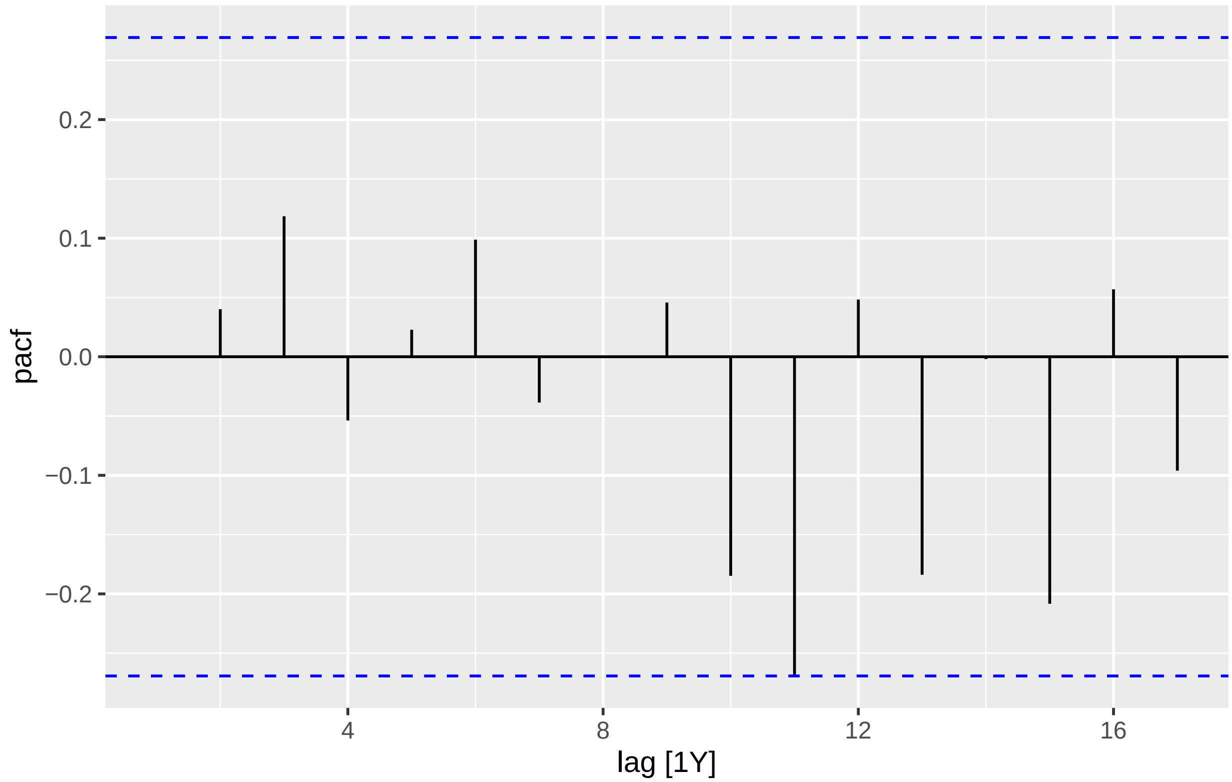
```
## # A mable: 1 x 2
## # Key:     Model name [1]
##   `Model name`                                         Orders
##   <chr>                                               <model>
## 1 tlb_sarima_policy <LM w/ ARIMA(2,1,2)(1,0,1)[12] errors>
```

```
tlb_perfect_resid <- augment(TLB_perfect)
tlb_perfect_resid |>
  ACF(.resid) |>
  autoplot() +
  labs(
    caption = stringr::str_wrap("tlb_sarima_policy_1 = ARIMA(log_TLB ~ small_family_push + big_family_pu
  )
```



tlb_sarima_policy_1 = ARIMA(log_TLB ~ small_family_push + big_family_push + pdq(2,1,2) + PDQ(1,0,1,period = 12))

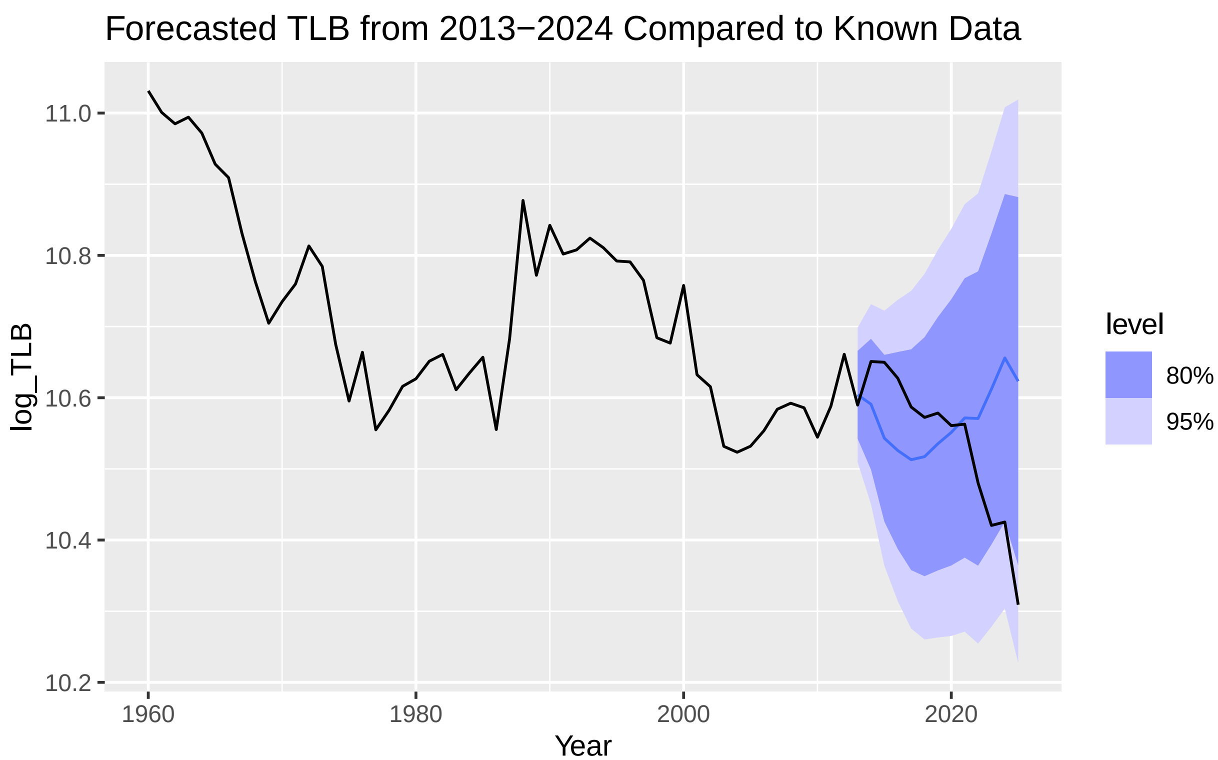
```
tlb_perfect_resid |>
  PACF(.resid) |>
  autoplot() +
  labs(
    caption = stringr::str_wrap("tlb_sarima_policy_1 = ARIMA(log_TLB ~ small_family_push + big_family_pu
  )
```



tlb_sarima_policy_1 = ARIMA(log_TLB ~ small_family_push + big_family_push + pdq(2,1,2) + PDQ(1,0,1,period = 12))

```
TLB_perfect_fc <- TLB_perfect |>
  forecast(new_data = singapore_future)
```

```
TLB_perfect_fc |>
  autoplot() +
  autolayer(singapore, log_TLB) +
  labs(
    title = "Forecasted TLB from 2013-2024 Compared to Known Data",
    caption = stringr::str_wrap("Shows the forecasted data from the SARIMA(2,1,2)(1,0,1)[12] model with
  ) +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0)
  )
```

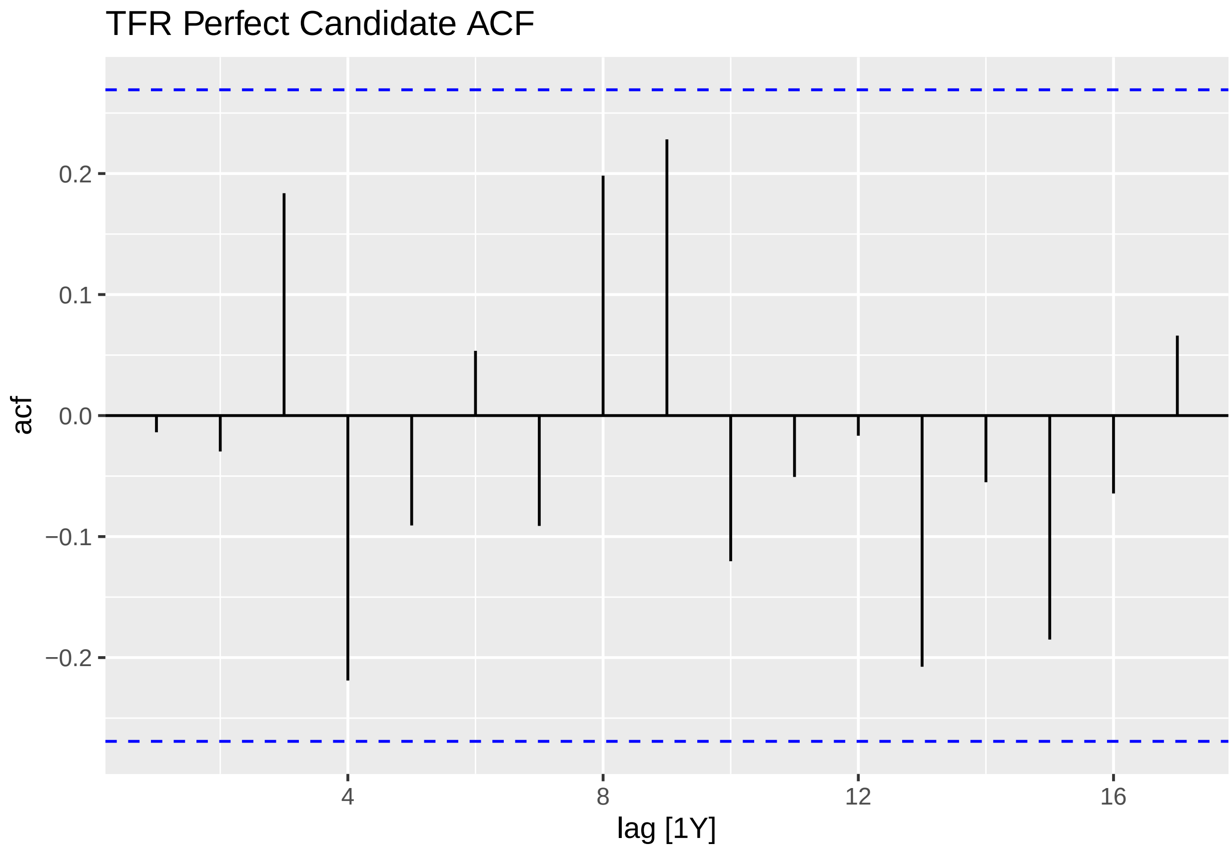


Shows the forecasted data from the SARIMA(2,1,2)(1,0,1)[12] model with government policies compares well to the known TLB data for the 2013−2024 period.

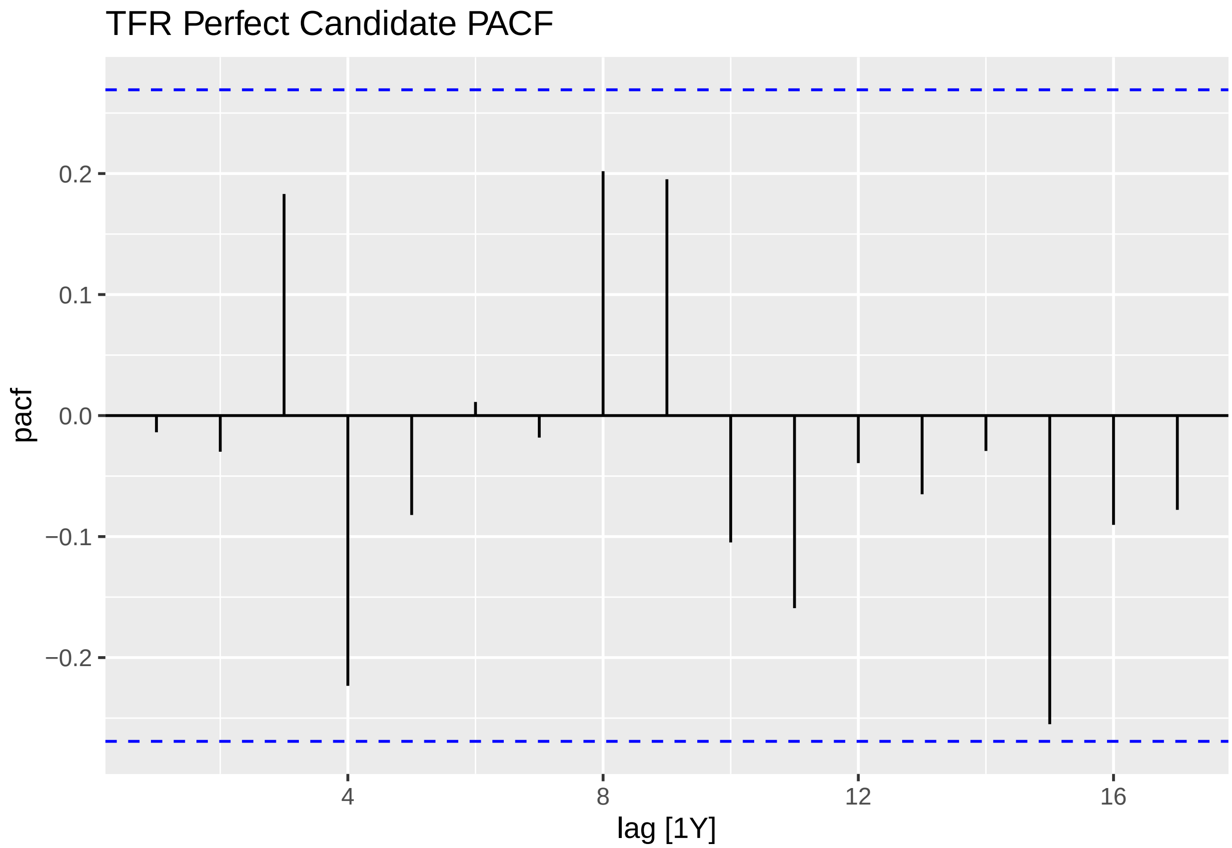
```
TFR_perfect <- singapore_train |>
  model(
    tfr_perfect = ARIMA(log_TFR ~ small_family_push + big_family_push + pdq(1,1,0) + PDQ(1,1,0, period =
  )

tfr_perfect_resid <- augment(TFR_perfect) |>
  filter(.model == "tfr_perfect")

tfr_perfect_resid |>
  ACF(.resid) |>
  autoplot() +
  labs(title = "TFR Perfect Candidate ACF")
```




```
tfr_perfect_resid |>
  PACF(.resid) |>
  autoplot() +
  labs(title = "TFR Perfect Candidate PACF")
```


TFR Perfect Candidate PACF
pacf
0.2
0.1
0.0
−0.1
−0.2
4
8
12
16
lag [1Y]